

Exploring Excess Return Generation through European Fixed Income Arbitrages

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Abstract:

1 Introduction

2 Literature Review

2.1 Fixed Income Arbitrage and the Limits to Arbitrage

The TIPS–Treasury basis documented by Fleckenstein et al. [2014] averaged 60 bps and peaked at 200 bps during the 2008–2009 crisis (Figure 1). Since then, the basis has effectively vanished (Figure 2), compressed by quantitative easing and broader institutional participation. In contrast, the Italian inflation-linked basis on BTP Italia persists to date—motivating our focus on the European market.

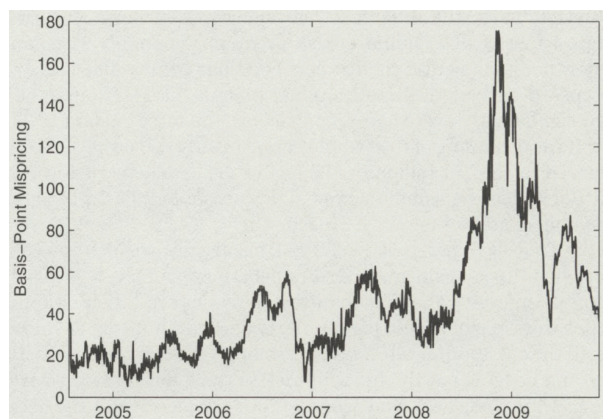


Figure 1
TIPS–Treasury basis, 2004–2009 (from Fleckenstein et al., 2014).

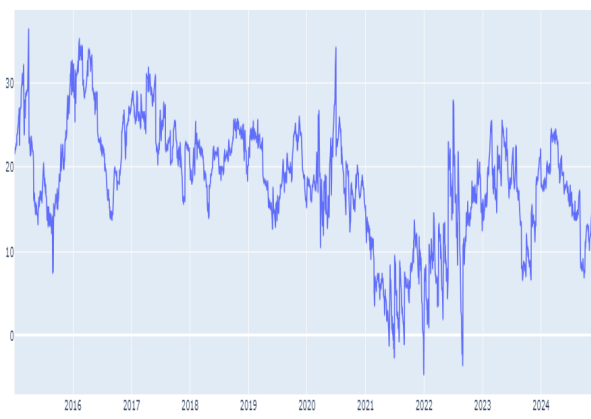


Figure 2
TIPS–Treasury basis, 2009–present. The mispricing has vanished.

2.2 Factor Models, Penalized Regression, and Principal Components

2.3 Slow-Moving Capital and Cross-Strategy Interdependencies

3 Data, Strategy Construction, and Performance

3.1 Data and Strategy Construction

Three strategies are constructed following the dynamic rolling fund approach of Duarte et al. [2007]: each trade is modelled as a separate fund, with entry triggered when the basis exceeds a

threshold and exit at convergence or maturity. Data are sourced from Bloomberg, ICPL, and ICE; samples begin in 2005 (CDS–Bond), 2008 (iTraxx), and 2014 (BTP Italia), all ending May 2025.

The **BTP Italia** strategy trades the inflation-linked basis between BTP Italia and nominal BTPs, where the inflation-linked leg is converted to a fixed-rate exposure via asset swap (ASW).

The **CDS–Bond Basis** strategy implements the negative basis trade (long bond, buy CDS protection) on investment-grade European corporates. The tradable universe is restricted to single-name CDS whose reference entity belongs to the current on-the-run iTraxx Main series; when a name exits the index, it is no longer traded, as we assume it loses liquidity. All CDS are centrally cleared, eliminating counterparty credit risk.

The **CDS Index Skew** strategy exploits the gap between the iTraxx index spread and the duration-weighted average of its single-name constituents. Only on-the-run series are traded, across four indices (Main, SnrFin, SubFin, Xover).

Entry/exit rules and thresholds are detailed in Appendix A.2.1.

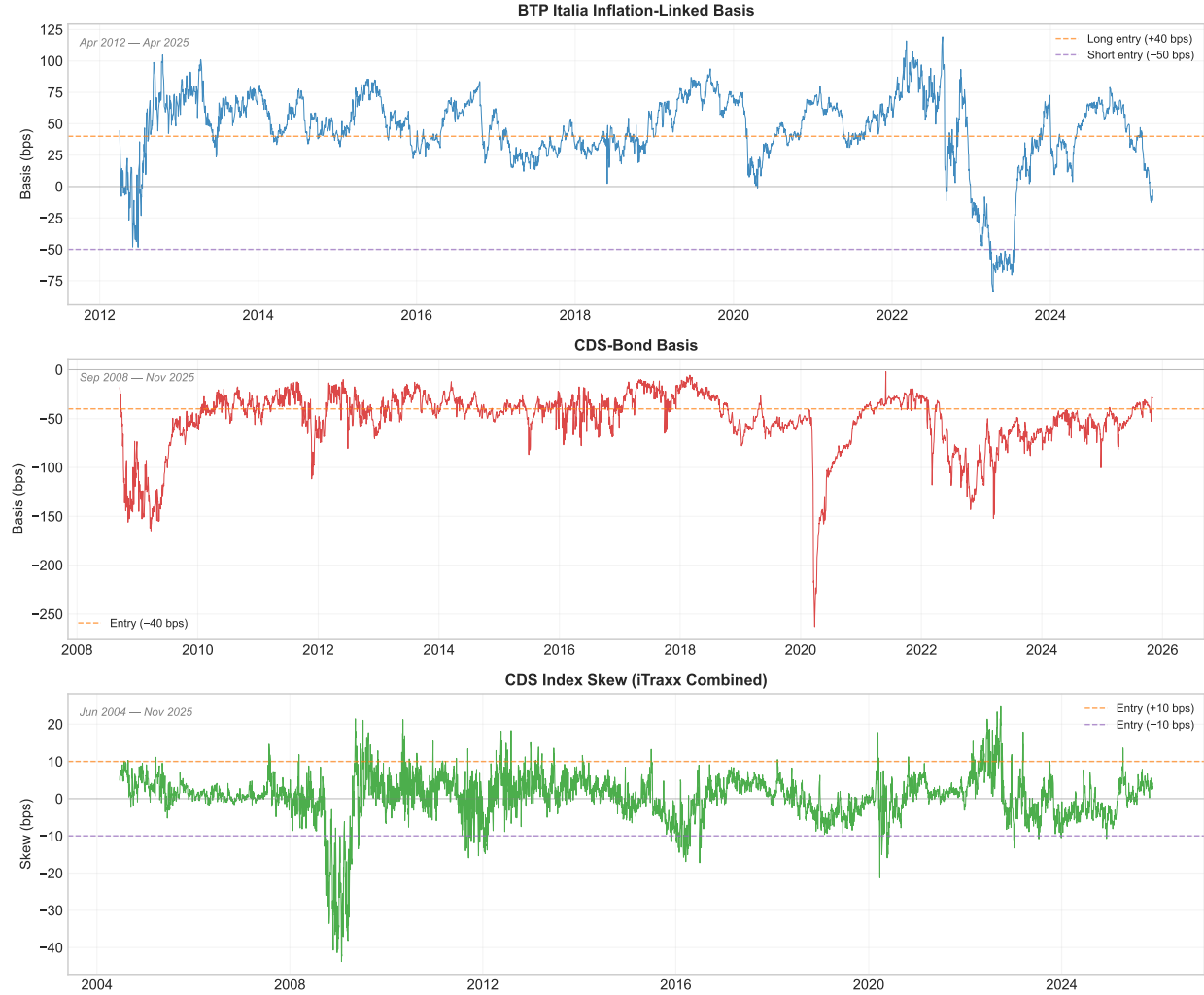


Figure 3

Market-Level Basis Time Series for Fixed Income Arbitrage Strategies. For BTP Italia, the basis is the cross-sectional mean of all outstanding issues with at least six months to maturity. For iTraxx Combined, it is the mean of the four on-the-run sub-index skews. For CDS–Bond Basis, it is the mean of the $N = 20$ most negative single-name bases in the iTraxx Main universe on each day (only bonds with negative basis are included). Dashed lines indicate entry thresholds.

3.2 Index Construction and Weighting Schemes

Returns are aggregated daily using both equal-weighted (Duarte et al., 2007) and signal-weighted (Rebonato and Ronzani, 2021) schemes, where SW weights each trade proportionally to the basis magnitude at entry. SW is adopted as baseline due to superior risk-adjusted performance (Appendix A.3). All returns are net of transaction costs computed from observed bid-ask spreads at

three regime-dependent tiers (LOW/MID/HIGH, determined by the level of iTraxx Main 5Y; Appendix A.2.3).

3.3 Performance Analysis

Table 1 reports summary statistics for the signal-weighted indices. All three strategies exhibit positive Sharpe ratios and moderate drawdowns, with the CDS–Bond basis showing the longest sample and iTraxx the highest annualised return.

Table 1
Summary Statistics for Fixed Income Arbitrage Strategies

Panel A: Basis Levels (bps)										
Strategy	Mean	Median	Std Dev	Min	Max	Skew	Kurt	AC(1)	P_{10}	P_{90}
<i>BTP Italia</i>	45.55	48.70	29.34	-83.9	119.3	-1.37	3.03	0.984	14.8	75.6
<i>CDS-Bond Basis</i>	7.03	10.06	18.44	-129.7	41.2	-1.01	1.90	0.944	-18.8	29.6
<i>iTraxx Combined</i>	0.40	1.05	6.45	-43.8	24.8	-1.74	8.42	0.872	-5.9	6.6
Panel B: Strategy Returns (%)										
Strategy	n	Mean (p.a.)	Std Dev (p.a.)	Min	Max	Skew	Kurt	% Neg	AC(1)	Sharpe
<i>BTP Italia</i>	3282	1.67	2.36	-1.60	1.77	0.60	22.85	48.9	-0.031	0.71
<i>CDS Bond Basis</i>	4169	5.37	3.15	-1.56	3.62	2.81	46.91	43.2	-0.029	1.71
<i>iTraxx Combined</i>	5018	1.96	2.58	-1.03	1.05	0.43	5.67	48.8	-0.231	0.76

Note: Panel A reports daily basis levels in basis points (Jan 2005–May 2025 for CDS–Bond, Oct 2008–May 2025 for iTraxx, Nov 2013–May 2025 for BTP Italia). For BTP Italia, the basis is the cross-sectional mean of all outstanding issues with at least six months to maturity. For iTraxx Combined, it is the mean of the four on-the-run sub-index skews (Main, SeniorFin, SubFin, Crossover). For CDS–Bond Basis, it is the cross-sectional median of all single-name bases in the iTraxx Main universe. P_{10} and P_{90} are the 10th and 90th percentiles of the daily time series. Panel B reports daily percentage returns, with *Mean* and *Std Dev* annualized ($\times 252$ and $\times \sqrt{252}$). *Skew* and *Kurt* denote skewness and excess kurtosis. *AC(1)* is the first-order autocorrelation. *Sharpe* is annualized. Sample periods vary by strategy.

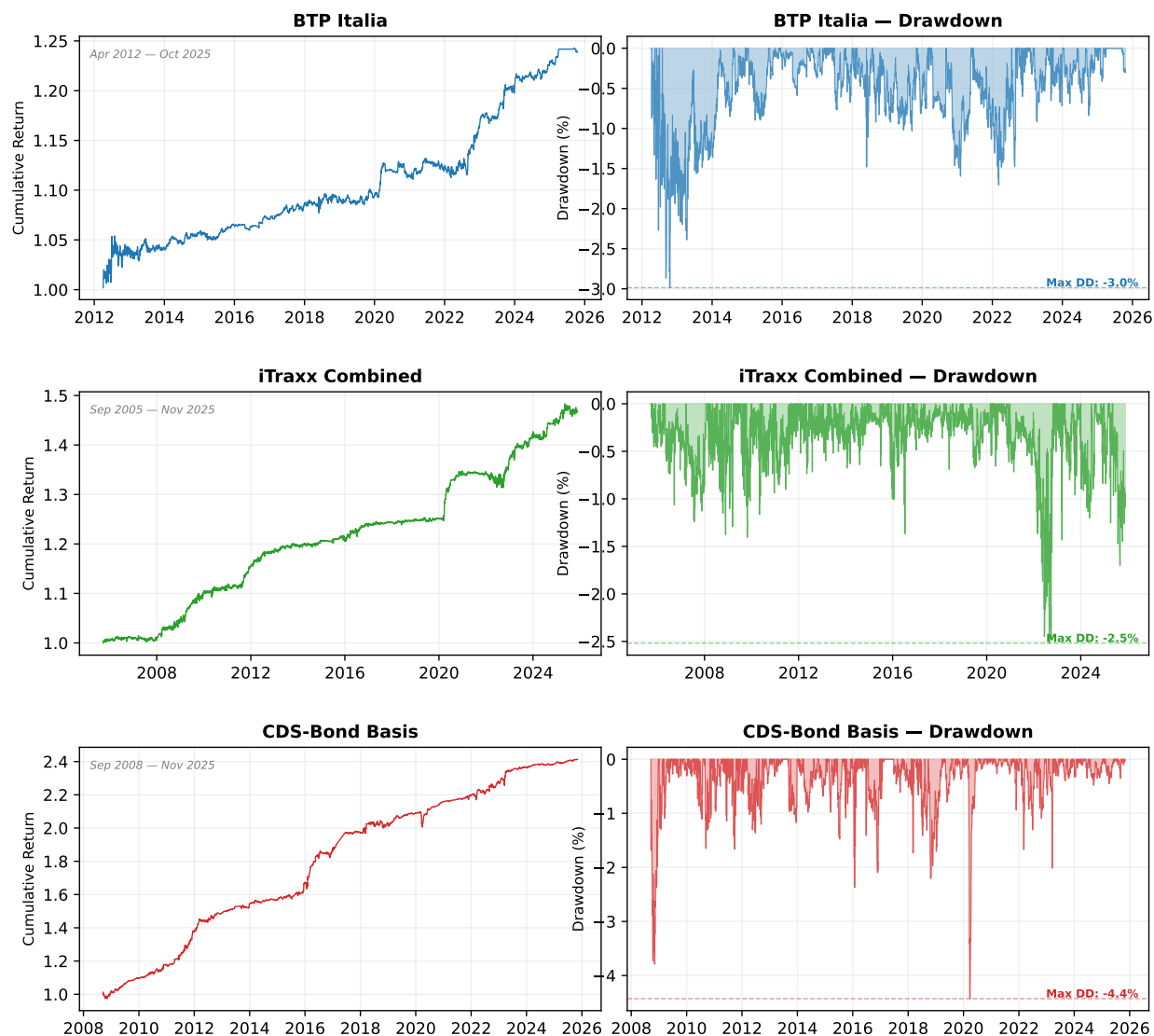


Figure 4
Cumulative Return Indices and Drawdowns of Fixed Income Arbitrage Strategies (Signal-Weighted).

Table 2 reports manipulation-proof performance measures [Goetzmann et al., 2007]. MPPM penalises strategies that inflate Sharpe ratios via dynamic trading or return smoothing; all three strategies retain significant risk-adjusted performance under this metric.

Table 2
Manipulation-Proof Performance Measure by Risk Aversion

Strategy	$\rho = 2$ (% p.a.)	$\rho = 3$ (% p.a.)	$\rho = 4$ (% p.a.)	Avg Ret (% p.a.)	Vol (% p.a.)	Sharpe	N
Panel A: EW							
BTP-Italia	1.31	1.30	1.28	1.34	1.69	0.79	163
CDS-Bond Basis	4.28	4.25	4.21	4.36	2.71	1.61	207
iTraxx <i>combined</i>	1.31	1.31	1.30	1.33	1.04	1.27	243
Panel B: SW							
BTP-Italia	1.56	1.55	1.54	1.59	1.62	0.98	163
CDS-Bond Basis	5.06	5.01	4.97	5.16	3.07	1.68	207
iTraxx <i>combined</i>	1.83	1.82	1.80	1.85	1.52	1.22	243

Note: MPPM is the Manipulation-Proof Performance Measure (Goetzmann et al. 2007). Computed from monthly returns compounded from daily. Risk-free is Euribor 1M (ACT/360), compounded to monthly.

Table 3 reports results for volatility-managed portfolios [Moreira and Muir, 2017], which scale exposure inversely to realised variance. Managed returns improve Sharpe ratios for two of the three strategies, consistent with time-varying risk premia rather than a static risk premium story.

Table 3
Moreira-Muir Volatility-Managed Portfolio Performance

Strategy	Buy-and-Hold			Vol-Managed			Alpha (% p.a.)	N
	Ret (% p.a.)	Vol (% p.a.)	Sharpe	Ret (% p.a.)	Vol (% p.a.)	Sharpe		
Panel A: EW								
BTP-Italia	1.27	1.67	0.76	0.21	1.67	0.13	-0.33	158
CDS-Bond Basis	4.27	2.63	1.63	2.83	2.63	1.08	+1.17	202
iTraxx <i>combined</i>	1.36	1.05	1.29	0.60	1.05	0.57	-0.01	242
Panel B: SW								
BTP-Italia	1.52	1.60	0.95	0.37	1.60	0.23	-0.22	158
CDS-Bond Basis	5.09	3.00	1.70	3.49	3.00	1.16	+1.68	202
iTraxx <i>combined</i>	1.91	1.53	1.25	0.84	1.53	0.55	-0.16	242

Note: Following Moreira and Muir (2017). The volatility-managed strategy scales positions inversely to realized variance: $f_{t+1}^\sigma = (c/\sigma_t^2) \times f_{t+1}$, where c matches buy-and-hold volatility. Alpha is from regressing VM returns on BH returns.

4 Benchmark Factor Models

4.1 Benchmark Specifications

We estimate three benchmark spanning regressions from the literature, each designed to capture different sources of systematic risk in fixed income strategies. Duarte et al. [2007] specify 10 factors spanning equity, bond market, and Treasury returns. Fung and Hsieh [2004] use 7 asset-based style factors including trend-following lookback straddles on bonds, currencies, and commodities. Brooks et al. [2020] decompose active fixed income performance into traditional risk premia (term, credit, emerging market debt, volatility, inflation-linked). All models are estimated via OLS with Newey–West HAC standard errors (6 lags).

Each benchmark is estimated in two panels. The first uses the original USD-denominated factors, replicating each framework exactly as specified by its authors. The second uses euro equivalents as a robustness check, since all three strategies operate in European markets. For the Fama–French factors—used directly in Duarte et al. [2007] and indirectly in Fung and Hsieh [2004]—we source the European portfolios from the Kenneth French data library; because these are reported in USD, we convert them to EUR following Glück et al. [2021]. For long factors (market excess return):

$$F_t^{\text{EUR}} = \frac{1 + F_t^{\text{USD}} + r_{f,t}^{\text{USD}}}{1 + r_t^{\text{USD/EUR}}} - 1 - r_{f,t}^{\text{EUR}}, \quad (1)$$

and for long-short factors (SMB, HML, UMD):

$$LS_t^{\text{EUR}} = \frac{LS_t^{\text{USD}}}{1 + r_t^{\text{USD/EUR}}}, \quad (2)$$

where F_t^{USD} and LS_t^{USD} are the original factors in US dollars, $r_{f,t}^{\text{USD}}$ and $r_{f,t}^{\text{EUR}}$ are the US and euro risk-free rates (Fama–French T-bill rate and Euribor 1M, respectively), and $r_t^{\text{USD/EUR}}$ is the USD/EUR exchange rate return. All remaining factors use native European equivalents: European bank stocks, euro-area corporate bond indices, and German government bonds (Duarte); euro-area term, credit, and inflation-linked indices (Brooks et al.); 10Y Bund yield and Baa–Bund credit spread (Fung &

Hsieh). The Fung & Hsieh trend-following lookback straddle factors (PTFSBD, PTFSFX, PTFS-COM) are currency-independent by construction (D. Hsieh, private communication) and are therefore identical in both panels. The complete factor list is reported in Appendix A.4; VIF diagnostics are in Appendix A.9.

4.2 Results and Discussion

Factor Model Regressions. Tables 4–6 report full regression results for the three benchmark specifications. Alpha is statistically significant across all frameworks and strategies, while R^2_{adj} remains low (5–15%), indicating that standard risk factors do not span returns.

Table 4
Duarte Factor Model Regressions

	α (%)	β_{Mkt}	β_{SMB}	β_{HML}	β_{UMD}	β_{R_S}	β_{R_I}	β_{R_B}	β_{R_2}	β_{R_5}	$\beta_{R_{10}}$	$\bar{R}^2$
Panel A: US Factors												
<i>BTP Italia</i>	2.05*** (4.41)	-0.07*** (-3.40)	-0.03** (-2.19)	-0.04** (-2.44)	0.00 (0.07)	0.04*** (3.40)	0.04 (0.55)	-0.08 (-0.94)	0.09 (0.35)	-0.05 (-0.31)	0.04 (0.54)	0.13
<i>CDS–Bond</i>	5.13*** (6.66)	-0.01 (-0.43)	0.00 (0.08)	0.01 (0.40)	-0.03* (-1.75)	-0.01 (-0.39)	0.26*** (3.32)	-0.06 (-1.59)	0.11 (0.31)	-0.46* (-1.84)	0.12 (1.08)	0.17
<i>Index Skew</i>	1.81*** (4.64)	0.02 (1.55)	-0.02* (-1.70)	-0.02 (-1.28)	-0.02* (-1.91)	-0.01 (-1.44)	-0.02 (-0.37)	-0.02 (-0.71)	0.34 (1.53)	-0.14 (-0.99)	0.05 (0.84)	0.05
Panel B: EUR Factors												
<i>BTP Italia</i>	1.65*** (3.36)	-0.05** (-1.99)	-0.06*** (-2.97)	-0.05** (-2.05)	0.02 (0.99)	0.02 (1.27)	-0.01 (-0.08)	0.03 (0.36)	0.05 (0.11)	-0.03 (-0.14)	-0.03 (-0.27)	0.11
<i>CDS–Bond</i>	5.14*** (5.59)	0.00 (0.08)	0.02 (0.53)	0.04 (1.13)	-0.03 (-1.38)	-0.03** (-2.01)	0.47*** (3.87)	-0.07* (-1.70)	-0.53 (-0.80)	0.15 (0.32)	-0.17 (-1.05)	0.14
<i>Index Skew</i>	2.26*** (4.64)	-0.02 (-1.20)	-0.04*** (-2.60)	-0.07*** (-2.58)	-0.03** (-2.27)	0.01 (1.16)	0.06 (1.00)	-0.02 (-0.99)	0.02 (0.05)	0.06 (0.26)	-0.07 (-0.90)	0.09

Note: Monthly excess returns regressed on the ten Duarte et al. (2007) equity and bond risk factors. α is annualized (multiplied by 12). t -statistics in parentheses (Newey–West HAC). Durbin–Watson statistics range from 1.83 to 2.13. The joint F -test rejects the null that all slope coefficients are zero at the 5% level for most specifications. Mkt : market excess return (Fama–French). R_S : bank stock excess return. R_I , R_B : A/BAA-rated industrial and bank bond excess returns. R_2 , R_5 , R_{10} : maturity-sorted Treasury portfolio excess returns. Sample: BTP Italia 162 months, CDS–Bond 205 months, Index Skew 241 months. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5
Fung & Hsieh Factor Model Regressions

	α (%)	$\beta_{S\&P}$	β_{SC-LC}	β_{BdOpt}	β_{FXOpt}	β_{ComOpt}	β_{10Y}	$\beta_{CredSpr}$	$\bar{R}^2$
Panel A: US Factors									
<i>BTP Italia</i>	1.80*** (3.64)	-0.02* (-1.96)	-0.02** (-2.08)	0.12 (0.72)	-0.01 (-0.03)	0.33 (1.05)	0.00 (0.01)	0.28 (1.18)	0.10
<i>CDS-Bond</i>	5.28*** (6.68)	-0.02 (-1.58)	0.01 (0.53)	0.09 (0.35)	-0.47 (-1.49)	-0.09 (-0.27)	-0.98*** (-3.98)	-1.73*** (-5.35)	0.18
<i>Index Skew</i>	1.69*** (4.32)	0.02 (1.46)	-0.02** (-2.09)	0.10 (0.48)	0.17 (0.89)	0.13 (0.48)	-0.07 (-0.57)	0.16 (0.78)	0.03
Panel B: EUR Factors									
<i>BTP Italia</i>	1.86*** (3.64)	-0.04*** (-2.99)	-0.06** (-2.40)	0.34* (1.80)	-0.08 (-0.39)	0.34 (0.92)	0.28 (1.44)	-0.53 (-1.13)	0.11
<i>CDS-Bond</i>	4.67*** (5.38)	-0.03 (-1.60)	0.04 (1.22)	0.21 (0.88)	-0.71* (-1.80)	-0.59 (-1.64)	-0.88** (-2.38)	-0.74 (-1.32)	0.11
<i>Index Skew</i>	3.48*** (6.05)	-0.00 (-0.28)	-0.07** (-2.02)	0.12 (0.53)	-0.13 (-0.56)	0.59* (1.83)	-0.34 (-1.37)	-0.12 (-0.38)	0.05

Note: Monthly excess returns regressed on the seven Fung & Hsieh (2004) hedge fund risk factors. α is annualized (multiplied by 12). t -statistics in parentheses (Newey–West HAC). Durbin–Watson statistics range from 1.49 to 1.95. The joint F -test rejects the null that all slope coefficients are zero at the 5% level for most specifications. S&P: S&P 500 excess return. SC–LC: Wilshire small-minus-large-cap equity spread. BdOpt, FXOpt, ComOpt: lookback-straddle returns on bond, currency, and commodity futures. 10Y: change in 10-year constant-maturity yield. CredSpr: change in Moody’s Baa minus 10-year Treasury spread. Sample: BTP Italia 162 months, CDS–Bond 195 months, Index Skew 195 months. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6
Active Fixed Income Factor Model Regressions

	α (%)	β_{Term}	$\beta_{\text{Global Term}}$	$\beta_{\text{Global Agg}}$	$\beta_{\text{Infl. Linkers}}$	$\beta_{\text{Corp Credit}}$	$\beta_{\text{Emerg Debt}}$	$\beta_{\text{Emerg Curr}}$	$\beta_{\text{UST Vol}}$	$\bar{R}^2$
Panel A: US Factors										
<i>BTP Italia</i>	1.88*** (3.65)	0.15 (0.98)	-0.27 (-0.88)	0.13 (0.31)	-0.02 (-0.82)	-0.14** (-2.37)	0.09* (1.72)	-0.02 (-0.91)	0.00 (0.15)	0.11
<i>CDS-Bond</i>	4.61*** (5.38)	0.01 (0.08)	-0.20 (-0.43)	0.42 (0.73)	-0.03 (-0.44)	-0.00 (-0.02)	0.07 (0.97)	0.02 (0.43)	0.01 (0.40)	0.10
<i>Index Skew</i>	1.57*** (4.36)	-0.05 (-0.45)	-0.10 (-0.43)	0.21 (0.68)	-0.01 (-0.17)	0.04 (1.08)	-0.07* (-1.82)	0.03 (1.49)	0.04** (2.00)	0.10
Panel B: EUR Factors										
<i>BTP Italia</i>	1.96*** (3.28)	-0.07 (-1.47)	-0.36 (-1.43)	0.51* (1.87)	-0.07** (-2.01)	-0.06 (-1.23)	-0.03 (-0.93)	0.03 (1.01)	0.00 (0.21)	0.01
<i>CDS-Bond</i>	5.07*** (5.87)	-0.02 (-0.17)	-0.05 (-0.12)	0.25 (0.70)	-0.02 (-0.24)	-0.08 (-1.52)	0.15** (2.21)	-0.02 (-0.31)	-0.02 (-1.30)	0.04
<i>Index Skew</i>	1.96*** (4.73)	0.03 (0.32)	-0.24 (-1.09)	0.26 (1.34)	-0.02 (-0.50)	0.01 (0.81)	-0.03 (-1.30)	-0.02 (-1.16)	-0.02* (-1.76)	0.01

Note: Monthly excess returns regressed on the eight Active Fixed Income premia of Brooks et al. (2020). α is annualized (multiplied by 12). t -statistics in parentheses (Newey–West HAC). Durbin–Watson statistics range from 1.78 to 2.22. The joint F -test rejects the null that all slope coefficients are zero at the 5% level for most specifications. Term: Treasury excess returns (US or Eurozone). Global Term/Agg: Bloomberg Global Treasury/Aggregate (hedged). Infl. Linkers: Bloomberg Global Inflation-Linked. Corp Credit: HY and leveraged loans in excess of duration-matched Treasuries. Emerg Debt/Curr: EM bond and FX baskets. UST Vol: delta-hedged straddles on 10Y Treasury futures. Sample: BTP Italia 163 months, CDS–Bond 207 months, Index Skew 243 months. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Cross-Model Comparison and Regime Analysis. Table 7 summarizes alpha across frameworks (Panel A) and decomposes it by stress regime (Panel B). Alpha is uniformly higher during HIGH stress months (iTraxx Main > 100 bps), consistent with Duffie [2010]: when funding constraints bind, arbitrage capital withdraws, mispricings widen, and alpha increases. This pattern motivates the cross-strategy analysis in Section 6.

Table 7
Alpha Estimates Across Benchmark Factor Models

Panel A: Full-sample alpha (% p.a.)

Strategy	Duarte et al. (2007)			Brooks et al. (2020)			Fung & Hsieh (2004)		
	α	t	N	α	t	N	α	t	N
<i>BTP Italia</i>	+1.65***	3.46	162	+1.93***	3.19	163	+1.86***	3.54	162
<i>iTraxx Combined</i>	+2.26***	4.58	241	+1.96***	4.50	243	+3.48***	5.31	195
<i>CDS–Bond Basis</i>	+5.14***	5.08	205	+5.07***	5.28	207	+4.67***	4.74	195

Panel B: Alpha by stress regime (% p.a.)

	Duarte et al. (2007)		Brooks et al. (2020)		Fung & Hsieh (2004)	
	HIGH	NORMAL	HIGH	NORMAL	HIGH	NORMAL
<i>BTP Italia</i>	+4.74** (2.17)	+1.65*** (3.29)	+4.13** (2.15)	+1.42*** (3.58)	+5.62*** (3.80)	+1.46*** (3.61)
<i>iTraxx Combined</i>	+2.71*** (3.32)	+1.98*** (3.05)	+4.38*** (4.48)	+1.38*** (3.01)	+7.25*** (4.71)	+2.61*** (4.91)
<i>CDS–Bond Basis</i>	+8.63*** (5.70)	+3.77*** (3.38)	+9.69*** (4.80)	+3.63*** (4.42)	+9.56*** (5.14)	+2.95*** (4.89)

Panel C: Ferson–Schadt (1996) conditional alpha (% p.a.)

	Duarte et al. (2007)		Brooks et al. (2020)		Fung & Hsieh (2004)	
	α_1	t -stat	α_1	t -stat	α_1	t -stat
<i>BTP Italia</i>	+1.09*	(1.66)	+1.50**	(2.48)	+1.39*	(1.95)
<i>iTraxx Combined</i>	+1.06***	(3.59)	+1.26***	(4.15)	+2.81***	(4.63)
<i>CDS–Bond Basis</i>	+2.87***	(2.97)	+2.90***	(3.29)	+3.31***	(3.12)

Note: Panel A reports annualized alpha (% p.a.) from monthly OLS regressions with Newey–West HAC standard errors using EUR factors. Panel B reports alpha separately for HIGH and NORMAL regimes, where HIGH denotes months in which the 5-year iTraxx Europe Main spread exceeds 100 bps. Panel C reports the stress-loading coefficient α_1 from the Ferson and Schadt [1996] conditional model $r_t = \alpha_0 + \alpha_1 z_t + \beta' X_t + \varepsilon_t$, where z_t is the standardized iTraxx Main 5Y spread. $\alpha_1 > 0$: alpha increases during funding stress. t -statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Conditional Alpha. Panel C of Table 7 formalises the regime effect using the Ferson and Schadt [1996] conditional model:

$$r_{i,t} = \alpha_0 + \alpha_1 z_t + \beta' \mathbf{X}_t + \varepsilon_t, \quad (3)$$

where $\mathbf{X}_t$ are the benchmark factors and z_t is the standardized iTraxx Main 5Y spread, proxying for intermediary funding stress. Unlike the binary regime split in Panel B, this specification allows alpha to vary continuously with market conditions. The coefficient $\alpha_1 > 0$ indicates that alpha increases during stress, consistent with Duffie [2010]: when intermediary capital is scarce, arbitrage positions offer a higher premium to compensate for the additional funding risk and balance-sheet cost of entering the trade.

Rolling Alpha. Figure 5 plots rolling 36-month alpha estimates for all three strategies under each benchmark specification. Alpha spikes visibly during stress episodes (shaded red), confirming that the regime effect documented in Table 7 is not driven by a single sub-period. Detailed sub-period splits and threshold robustness for each framework are reported in Appendix A.7.

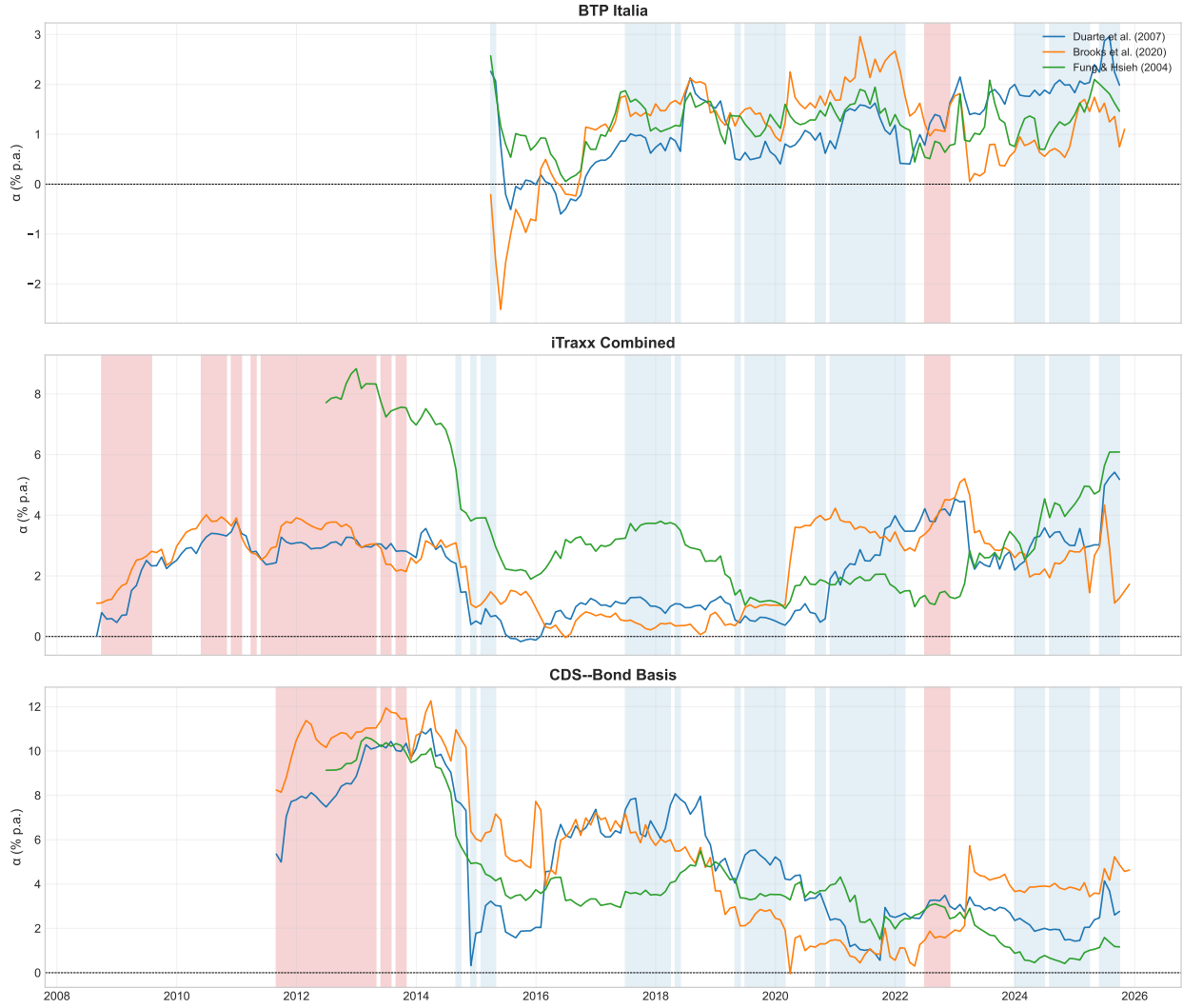


Figure 5

Rolling 36-Month Alpha Across Benchmark Factor Models. Each panel shows one strategy. Lines represent rolling alpha under Duarte et al. (2007, blue), Brooks et al. (2020, orange), and Fung & Hsieh (2004, green). Red shading: HIGH stress regime (iTraxx Main 5Y > 100 bps).

5 Factor Selection and Spanning Regressions

5.1 Principal Component Analysis

5.1.1 Rolling PCA Methodology

We extract principal components from a rolling window of $W = 24$ months over 80+ standardised candidate factors (Ludvigson and Ng, 2009). $K = 8$ components are retained (approximately 80%

cumulative variance; Figure 6). Both predictive ($PC_t \rightarrow R_{t+1}$) and contemporaneous ($PC_t \rightarrow R_t$) timing are tested. Spanning regressions are estimated by OLS with Newey–West HAC standard errors. Robustness across (W, K) grids is reported in Appendix A.5.

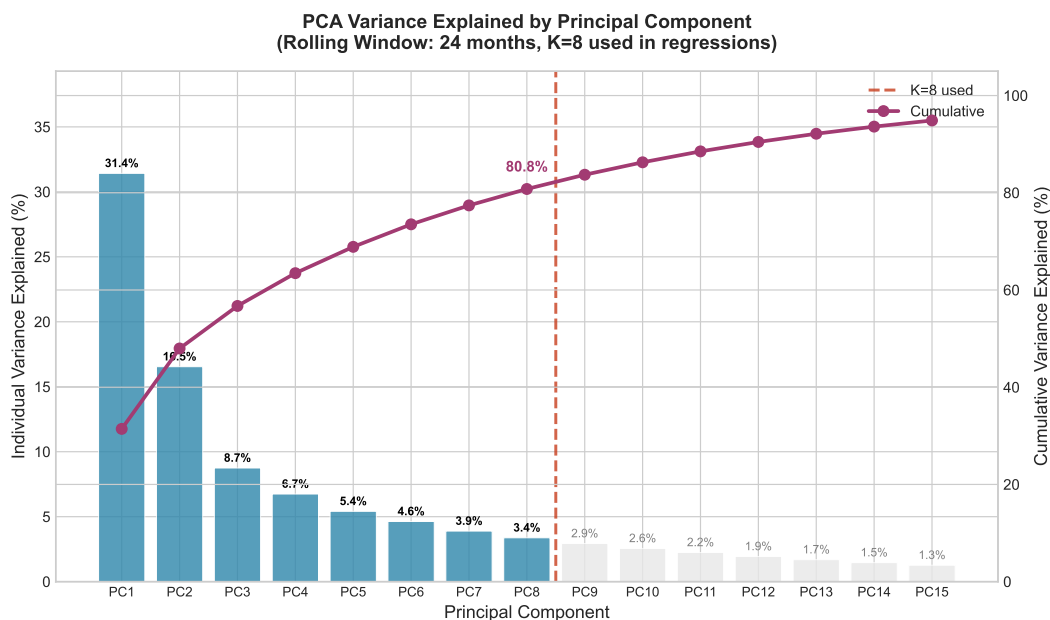


Figure 6

Scree Plot: Cumulative Variance Explained by Principal Components. Dashed line indicates 80% threshold. Baseline specification uses $K = 8$ components.

5.1.2 Results

Table 8 reports spanning regression results. Alpha is statistically significant for all three strategies under both timing conventions, while $\bar{R}^2$ remains moderate, indicating that diffuse systematic risk does not span strategy returns.

Table 8
PCA Spanning Regressions

	α (%)	β_{PC1}	β_{PC2}	β_{PC3}	β_{PC4}	β_{PC5}	β_{PC6}	β_{PC7}	β_{PC8}	$\bar{R}^2$	N
Panel A: Predictive ($PC_t \rightarrow R_{t+1}$)											
<i>BTP Italia</i>	1.68*** (4.05)	-0.012 (-1.13)	-0.001 (-0.07)	-0.026 (-1.46)	-0.019 (-0.61)	-0.012 (-0.52)	-0.028 (-1.46)	-0.001 (-0.02)	-0.000 (-0.02)	-0.005	156
<i>CDS-Bond</i>	5.43*** (5.91)	0.018 (1.23)	0.002 (0.09)	0.068* (1.69)	0.059 (1.17)	-0.060** (-2.57)	0.011 (0.50)	-0.003 (-0.08)	0.089*** (3.03)	0.050	196
<i>Index Skew</i>	2.07*** (5.38)	-0.011 (-1.20)	-0.009 (-0.95)	-0.002 (-0.13)	0.013 (0.78)	0.023 (0.89)	0.015 (0.62)	0.026 (1.47)	0.042** (2.11)	0.041	208
Panel B: Contemporaneous ($PC_t \rightarrow R_t$)											
<i>BTP Italia</i>	1.61*** (3.48)	-0.025** (-1.99)	-0.010 (-0.97)	-0.028 (-1.20)	0.017 (1.05)	0.008 (0.41)	0.003 (0.13)	-0.013 (-0.50)	0.010 (0.60)	0.054	157
<i>CDS-Bond</i>	5.87*** (5.74)	0.035*** (3.07)	0.010 (0.52)	0.067* (1.95)	-0.016 (-0.54)	-0.032 (-0.80)	-0.079** (-2.39)	0.076** (2.02)	0.051 (1.03)	0.120	197
<i>Index Skew</i>	2.21*** (5.01)	0.001 (0.16)	-0.003 (-0.29)	0.018 (0.90)	0.042 (1.64)	-0.007 (-0.54)	0.012 (0.65)	-0.039** (-2.19)	0.018 (1.21)	0.049	209

Note: Monthly strategy excess returns regressed on the first 8 principal components extracted from 24-month rolling windows. α is annualized (multiplied by 12). t -statistics in parentheses (Newey–West HAC). The first 8 PCs explain on average 80.8% of factor variance (min 75.7%, max 87.8%). Sample: BTP Italia 156 months, CDS–Bond 196 months, Index Skew 208 months. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 9
Top Factor Loadings by Principal Component

PC1		PC2		PC3	
Factor	Loading	Factor	Loading	Factor	Loading
EMERG_DEBT	+0.088	GLOBAL_AGG	+0.120	SS2Y	+0.062
CORP_CREDIT	+0.087	GLOBAL_TERM	+0.109	SS5Y	+0.053
ITRX_MAIN	-0.080	RI_EU	+0.109	PB_EU_CDS_5Y	+0.046
RS_EU	+0.079	$\Delta 10Y_YIELD_EU$	-0.104	ΔVIX	-0.042
DEF_US	+0.079	R2_EU	+0.103	PB_EU_CDS_1Y	+0.040
MKT_EU	+0.079	TERM_EU	+0.103	EP_SVIX_1M	+0.040

Note: Loadings are time-series averages of the eigenvector entries across all 24-month rolling windows. For each principal component, the 6 factors with the largest absolute loading are reported. Sign indicates the direction of the relationship.

5.2 Adaptive Elastic Net

5.2.1 Candidate Factor Universe and Preprocessing

The same 80+ candidate factors used in the PCA analysis (Appendix A.4.1) are preprocessed for penalised regression: stationarity is verified via ADF tests, factors are standardised to unit L_2 -norm [Zou and Zhang, 2009], and near-duplicate pairs ($|\rho| > 0.95$) are removed. After cleaning, approximately 75+ factors remain per strategy.

5.2.2 Estimation and Tuning

The Adaptive Elastic Net [Zou and Zhang, 2009] proceeds in two stages. Stage 1 estimates a standard Elastic Net:

$$\hat{\beta}^{EN} = \arg \min_{\beta} \left\{ \frac{1}{T} \sum_{t=1}^T (r_t - \alpha - \beta' f_t)^2 + \lambda_2 \|\beta\|_2^2 + \lambda_1 \|\beta\|_1 \right\}.$$

Stage 2 constructs adaptive weights $w_j = |\hat{\beta}_j^{EN}|^{-\gamma}$ ($\gamma = 1$) and re-solves with factor-specific ℓ_1 penalties:

$$\hat{\beta}^{AEN} = \arg \min_{\beta} \left\{ \frac{1}{T} \sum_{t=1}^T (r_t - \alpha - \beta' f_t)^2 + \lambda_2 \|\beta\|_2^2 + \lambda_1 \sum_{j=1}^p w_j |\beta_j| \right\}.$$

Both stages use the same λ_2 ; (λ_1, λ_2) are selected by minimising the Hannan–Quinn criterion (HQC). The solution is obtained via coordinate descent [Friedman et al., 2010].

5.2.3 Stability Selection

To address selection uncertainty, we apply stability selection [Meinshausen and Bühlmann, 2010] adapted for time-series dependence: $B = 100$ stationary bootstrap resamples (expected block length 9 months) are drawn, and the full AEN is re-run on each. A factor is retained only if its selection frequency $\hat{\pi}_j \geq 60\%$. The expected number of false selections is bounded by $q^2/[p(2\pi^* - 1)]$. Cluster-level frequency aggregation corrects for vote-splitting among correlated factors

5.2.4 *Post-Selection OLS Results*

Once the stable set $\hat{S}_i$ is identified, we estimate unrestricted OLS on the original (un-standardised) factors:

$$r_{i,t} = \alpha_i + \beta_i' f_{\hat{S}_i,t} + \varepsilon_{i,t}$$

with Newey–West HAC standard errors (6 lags).

Table 10 reports results. Alpha is statistically significant for all three strategies. The selected factors are economically interpretable and span several categories: liquidity and funding frictions, credit risk premia, equity style factors, and interest rate term structure variables. Each strategy loads on a distinct combination, consistent with the different market microstructures and investor bases involved. Compared with the three benchmark frameworks (Section 4) and the PCA spanning regressions, the AEN-selected factors deliver higher adjusted $\bar{R}^2$ while preserving significant alpha, confirming that these targeted risk drivers capture strategy-specific exposures that diffuse models miss.

Table 10
Adaptive Elastic Net: Post-Selection OLS on Stable Factors

<i>Panel A: BTP Italia</i>		
	Coefficient	<i>t</i> -stat
α (ann. %)	+1.39***	3.34
SMB_EU	−0.044**	−2.30
ILLIQ	+0.638**	2.50
SS10Y	−0.021***	−3.26
UMD_EU	+0.030**	1.96
CDX_IG	+0.009	1.60
LIBOR_OIS	+0.490*	1.90
$T = 157, k = 6, \bar{R}^2 = 0.2173, DW = 2.0351$		
<i>Panel B: CDS–Bond Basis</i>		
α (ann. %)	+3.94***	3.66
ΔUF	−4.271***	−2.61
RI_EU	+0.160***	3.27
EMERG_FX	+0.061**	1.99
PB_EU_CDS_1Y	+0.011**	2.02
SS5Y	+0.018**	2.33
CRED_SPR_US	−0.510**	−2.20
EP_SVIX_1M	+3.245***	2.82
PTFSFX	−0.450*	−1.74
$T = 197, k = 8, \bar{R}^2 = 0.2670, DW = 2.0233$		
<i>Panel C: iTraxx Combined</i>		
α (ann. %)	+0.47	1.04
EP_SVIX_1M	+3.306***	2.89
LIBOR_OIS	+0.484**	2.21
$\Delta V2X$	−0.017**	−2.55
ΔUR	+3.015**	2.35
$\Delta FAILS_PCT_TSY$	+0.402***	3.20
$T = 237, k = 5, \bar{R}^2 = 0.2359, DW = 2.2489$		

Note: Post-selection OLS on factors identified by bootstrap stability selection ($\hat{\Pi}_j \geq 0.80$). Factors are in original (un-standardized) units. α is annualized (% p.a.). Coefficients rounded to 3 decimals. *t*-statistics based on Newey–West HAC standard errors (6 lags). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Factor definitions are provided in Table 22.

5.3 Conditional Alpha

Alpha increases during periods of financial stress across all strategies and both factor-selection methods (Table 11). This is consistent with the limits-to-arbitrage mechanism [Duffie, 2010, Mitchell et al., 2007]: when funding constraints bind, capital retreats, mispricings widen, and alpha grows. Detailed Ferson–Schadt and threshold robustness results are in Appendix A.5.3 and A.6.3.

Table 11
Alpha by Stress Regime: PCA and AEN

Strategy	PCA		AEN	
	HIGH	NORMAL	HIGH	NORMAL
<i>BTP Italia</i>	+5.37** (2.24)	+1.28*** (4.21)	+2.71 (1.43)	+1.40*** (3.64)
<i>CDS–Bond Basis</i>	+9.13*** (3.77)	+4.84*** (4.43)	+8.17*** (4.62)	+4.15*** (4.41)
<i>iTraxx Combined</i>	+3.38*** (4.36)	+1.82*** (4.22)	+1.51** (2.03)	-0.94 (-1.45)

Note: Annualized alpha (% p.a.) from monthly OLS regressions with Newey–West HAC standard errors. PCA: spanning regression on 8 rolling principal components. AEN: post-selection OLS on bootstrap-stable factors. HIGH denotes months in which the 5-year iTraxx Europe Main spread exceeds 100 bps. *t*-statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.4 Discussion, Method Comparison, and Robustness

Table 12 compares alpha across estimation methods: AEN, Elastic Net, LASSO, Adaptive LASSO, and ALASSO-Chen. Alpha is robust to the choice of method. AEN is preferred for its oracle property, its ability to handle collinearity, and the stability selection step that reduces post-selection bias. Sensitivity to the correlation threshold ρ and the adaptive exponent γ is reported in Appendix A.6.1.

Table 12
Factor Selection Method Comparison

Method	BTP Italia			CDS–Bond Basis			iTraxx Combined		
	k	α	$\bar{R}^2$	k	α	$\bar{R}^2$	k	α	$\bar{R}^2$
Lasso	7	+1.39*** (3.39)	0.2203	11	+3.91*** (3.44)	0.2594	13	+0.67 (1.58)	0.3016
Elastic Net	7	+1.39*** (3.39)	0.2203	11	+3.91*** (3.44)	0.2594	13	+0.67 (1.58)	0.3016
Adaptive Lasso	7	+1.39*** (3.39)	0.2203	8	+3.94*** (3.66)	0.2670	11	+0.70 (1.63)	0.3056
AEN	6	+1.39*** (3.34)	0.2173	8	+3.94*** (3.66)	0.2670	11	+0.70 (1.63)	0.3056
ALASSO-Chen	51	+0.98 (1.62)	0.4818	29	+4.97*** (2.90)	0.3509	19	+0.74* (1.68)	0.3571

Note: α is annualized (% p.a.) from post-selection OLS. k denotes the number of selected factors. t -statistics (Newey–West HAC) in parentheses. AEN, LASSO, and Elastic Net use the same tuning criterion (HQC). Adaptive LASSO (Chen) uses AIC per Chen and Chen (2008). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6 Cross-Strategy Interdependencies

6.1 Correlations, Co-Widening, and Persistence

This section examines unconditional and regime-dependent co-movement across the three arbitrage strategies, using mispricing magnitudes $m_{i,t}$ (basis levels) rather than returns, following Duffie [2010].

Mispricing Construction. We construct $m_{i,t}$ as the absolute basis level for each strategy, sign-adjusted so that $\Delta m > 0$ denotes widening (increased mispricing) for all three. BTP Italia: cross-sectional mean with maturity filter. CDS-Bond: top negative-basis names. iTraxx Skew: mean of on-the-run series. Regime definition uses iTraxx Main 5Y with fixed thresholds (see Section 3).

Unconditional and Regime-Dependent Correlations. Table 13 reports Pearson correlations for both strategy returns (Panel A) and mispricing changes Δm (Panel B), unconditionally and across

three stress regimes (LOW/MEDIUM/HIGH, iTraxx Main 5Y thresholds at 60 and 120 bps). Correlations display clear monotonicity: the corporate credit pair (CDS–Bond/iTraxx) becomes more positively correlated as stress increases, while pairs involving BTP Italia become progressively more negative. This two-cluster pattern is interpreted in Section 6.4. Figure 7 visualizes the regime contrast.

Table 13
Unconditional and Regime-Dependent Correlations

Panel A: Strategy Returns

Pair	Unconditional		By Regime		
	Pearson	<i>t</i> -stat	LOW	MED	HIGH
CDS–Bond Basis vs BTP Italia	−0.183	−1.51	−0.077	−0.242**	−0.625*
CDS–Bond Basis vs iTraxx Combined	−0.019	−0.16	−0.117	−0.021	−0.255
BTP Italia vs iTraxx Combined	0.217***	2.59	0.199	0.164	0.417

Panel B: Mispricing Changes (Δm)

Pair	Unconditional		By Regime		
	Pearson	<i>t</i> -stat	LOW	MED	HIGH
CDS–Bond Basis vs BTP Italia	−0.260***	−2.89	−0.173	−0.291***	−0.345
CDS–Bond Basis vs iTraxx Combined	0.182*	1.65	0.099	0.098	0.380*
BTP Italia vs iTraxx Combined	−0.152**	−1.99	0.092	−0.169	−0.530

Note: Pearson correlations (unconditional) with HAC *t*-statistics (Newey–West). Regime columns: Pearson correlations in LOW, MEDIUM, and HIGH stress (iTraxx Main 5Y thresholds). Stars on unconditional: HAC inference. Stars on regime correlations: two-sided *t*-test with $n - 2$ degrees of freedom. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

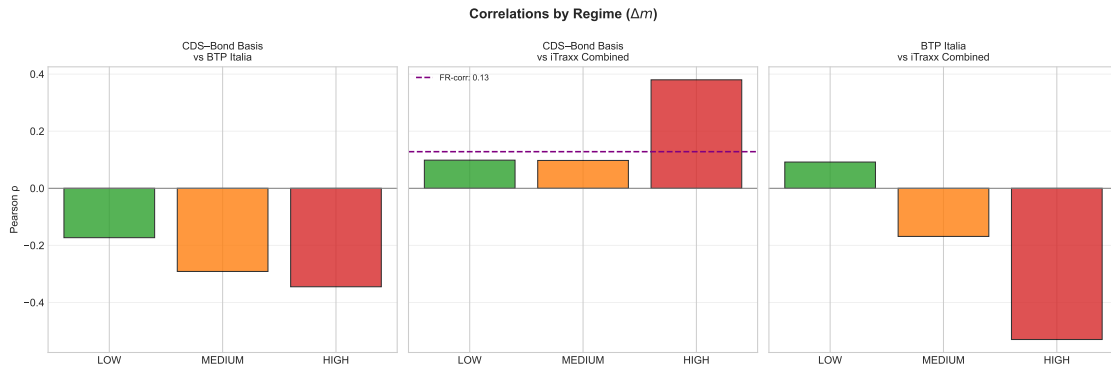


Figure 7

Pairwise Correlations by Stress Regime. Bars show Pearson correlations of Δm in LOW, MEDIUM, and HIGH stress regimes defined by iTraxx Main 5Y thresholds (60/120 bps).

6.2 Spanning Regressions Across Strategies

Cross-Strategy Spanning. Following Fleckenstein et al. [2014], we estimate spanning regressions of the form

$$\Delta m_{i,t} = \alpha + \sum_{j \neq i} \sum_{l=0}^L \beta_{j,l} \Delta m_{j,t-l} + \varepsilon_t, \quad (4)$$

where $\Delta m_{i,t}$ is the monthly change in mispricing for strategy i and $L = 2$ lags. Inference uses Newey–West HAC standard errors. Table 14 reports the results. Significant contemporaneous coefficients indicate that mispricing changes co-move beyond what common macro factors would predict. Significant lagged coefficients point to lead-lag transmission across strategies, consistent with slow capital reallocation: the more liquid market (iTraxx index) adjusts first, and the effect propagates to the less liquid markets (CDS–Bond, BTP Italia) with a delay of one to two months. The $\bar{R}^2$ is highest for BTP Italia, indicating that the corporate credit cluster has meaningful predictive power for the inflation-linked basis despite the two-cluster segmentation.

Table 14
Cross-Strategy Spanning Regressions (Δm)

Dependent	Regressor	β	t -stat	p	$\bar{R}^2$
CDS–Bond Basis	BTP Italia_L0	−0.232*	-1.91	0.0560	0.0973
	BTP Italia_L1	−0.089	-0.87	0.3847	
	BTP Italia_L2	−0.069	-1.04	0.2980	
	iTraxx Combined_L0	+1.270	1.26	0.2060	
	iTraxx Combined_L1	+1.344*	1.88	0.0597	
	iTraxx Combined_L2	+1.081*	1.72	0.0858	
BTP Italia	iTraxx Combined_L0	−1.264**	-2.52	0.0116	0.1328
	iTraxx Combined_L1	−1.558**	-1.98	0.0472	
	iTraxx Combined_L2	+0.407	0.47	0.6406	
	CDS–Bond Basis_L0	−0.220***	-2.64	0.0083	
	CDS–Bond Basis_L1	+0.171**	2.21	0.0271	
	CDS–Bond Basis_L2	−0.037	-0.55	0.5789	
iTraxx Combined	BTP Italia_L0	−0.013	-1.18	0.2380	0.0545
	BTP Italia_L1	+0.033	1.60	0.1089	
	BTP Italia_L2	+0.022**	2.04	0.0415	
	CDS–Bond Basis_L0	+0.016	1.08	0.2819	
	CDS–Bond Basis_L1	+0.010	0.86	0.3916	
	CDS–Bond Basis_L2	−0.009	-0.71	0.4774	

Note: $\Delta m_{i,t} = \alpha + \beta \Delta m_{j,t} + \varepsilon_t$. Newey–West HAC standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Stress Amplification. Duffie [2010] predicts that cross-strategy transmission should intensify when intermediary capital is under pressure. Table 15 tests this via interaction regressions:

$$\Delta m_{i,t} = \beta_0 \Delta m_{j,t} + \beta_1 (\Delta m_{j,t} \times z_t) + \gamma z_t + \varepsilon_t, \quad (5)$$

where z_t is the standardized iTraxx Main 5Y spread, proxying for funding stress. The coefficient β_1 captures the *incremental* transmission during stress: $\beta_1 > 0$ means that a given shock to Δm_j has a larger effect on Δm_i when intermediary balance sheets are strained. Results confirm the prediction for the corporate credit cluster, while pairs involving BTP Italia exhibit significant *negative* amplification—stress widens the gap between the institutional and retail segments, consistent with the two-cluster structure documented above.

Table 15
Stress Amplification: Interaction Regressions

Dependent	Variable	Coefficient	<i>t</i> -stat	<i>p</i>	$\bar{R}^2$
CDS–Bond Basis	Δm_j (BTP Italia)	−0.229	−1.60	0.110	0.0937
	$\Delta m_j \times z_t$	−0.061	−0.65	0.518	
	z_t (stress)	+3.159**	2.49	0.013	
BTP Italia	Δm_j (CDS–Bond Basis)	−0.223**	−2.32	0.021	0.0744
	$\Delta m_j \times z_t$	−0.156	−1.18	0.239	
	z_t (stress)	+0.665	0.34	0.736	
CDS–Bond Basis	Δm_j (iTraxx Combined)	+0.746	1.06	0.290	0.0585
	$\Delta m_j \times z_t$	+0.186	0.35	0.728	
	z_t (stress)	+2.399***	2.72	0.006	
iTraxx Combined	Δm_j (CDS–Bond Basis)	+0.017	1.25	0.210	0.0460
	$\Delta m_j \times z_t$	+0.023	0.93	0.354	
	z_t (stress)	+0.022	0.09	0.926	
BTP Italia	Δm_j (iTraxx Combined)	−0.521	−1.01	0.314	0.0450
	$\Delta m_j \times z_t$	−1.265**	−2.26	0.024	
	z_t (stress)	−0.071	−0.04	0.969	
iTraxx Combined	Δm_j (BTP Italia)	−0.013	−1.25	0.211	0.0482
	$\Delta m_j \times z_t$	−0.027***	−3.11	0.002	
	z_t (stress)	+0.184	0.89	0.374	

Note: $\Delta m_{i,t} = \beta_0 \Delta m_{j,t} + \beta_1 (\Delta m_{j,t} \times z_t) + \gamma z_t + \varepsilon_t$, where z_t is the standardized iTraxx Main 5Y spread. $\beta_1 > 0$: interdependence increases under stress. Newey–West HAC standard errors.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Common Factor Structure. Table 16 reports the regression of PC1 (first principal component of the three Δm series) on intermediary stress proxies, following the logic of Duffie [2010]: if the common factor driving correlated mispricings reflects intermediary balance-sheet constraints, it should be explained by variables that capture dealer capital, funding costs, and market illiquidity. Panel A selects one proxy per channel from the intermediary asset pricing literature: the He et al. [2017] capital ratio (HKM_IC), the Libor–OIS spread as a measure of global dealer funding cost, European prime broker CDS spreads as a direct gauge of dealer health, and the Amihud and Mendelson [2015] illiquidity measure. The majority of proxies are significant with signs consistent with theory: a decline in intermediary capital, a rise in funding costs, or an increase in market

illiquidity all predict wider mispricings. Panel B confirms these findings with an extended set of proxies, providing direct evidence for the slow-moving capital channel.

Table 16
Common Factor (PC1) and Funding Stress

Factor	Coefficient	<i>t</i> -stat	<i>p</i> -value
Panel A: Core Intermediary Proxies			
$T = 149, \bar{R}^2 = 0.231$			
HKM_IC	−4.292**	−2.23	0.026
LIBOR_OIS	+3.336***	3.07	0.002
PB_EU_CDS_5Y	−0.018**	−1.97	0.049
ILLIQ	+1.730***	2.79	0.005
Panel B: Extended Proxy Set			
$T = 149, \bar{R}^2 = 0.235$			
HKM_IC	−5.289***	−2.85	0.004
LIBOR_OIS	+3.247***	3.02	0.003
PB_EU_CDS_5Y	−0.011	−1.04	0.299
ILLIQ	+2.007***	2.95	0.003
GC-REPO_T-BILL	−0.130	−0.35	0.727
BFCI_EU	+0.527	1.54	0.124
ΔFAILS_PCT_TSY	−5.490	−0.94	0.348

Note: OLS regression of the first principal component of the three Δm series on intermediary stress proxies. Panel A: one proxy per Duffie channel (intermediary capital, funding cost, dealer health, market illiquidity). Panel B: all available funding and liquidity proxies. Significant coefficients indicate that the common factor driving correlated mispricings is linked to intermediary balance-sheet conditions [Duffie, 2010, He et al., 2017]. Newey–West HAC standard errors.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.3 VAR Analysis and Impulse Responses

Granger Causality. The results reveal a clear transmission hierarchy: the iTraxx index — the most liquid instrument in the system — Granger-causes BTP Italia, and CDS–Bond Basis also Granger-causes BTP Italia. This is consistent with a liquidity pecking order in which funding shocks first affect the standardised, exchange-traded credit index, then propagate to the less liquid cash-CDS basis and finally to the retail-dominated inflation-linked market.

Generalized Impulse Responses. Figure 8 plots Generalized IRFs [Pesaran and Shin, 1998], which are invariant to variable ordering. Cross-responses confirm the Granger causality findings: a shock to iTraxx Δm generates a significant negative response in BTP Italia that persists for approximately two to three months, consistent with the two-cluster structure. Own-responses on the diagonal reveal that CDS–Bond shocks are the most persistent (half-life of approximately four months), while iTraxx shocks dissipate within two months — consistent with the liquidity hierarchy.

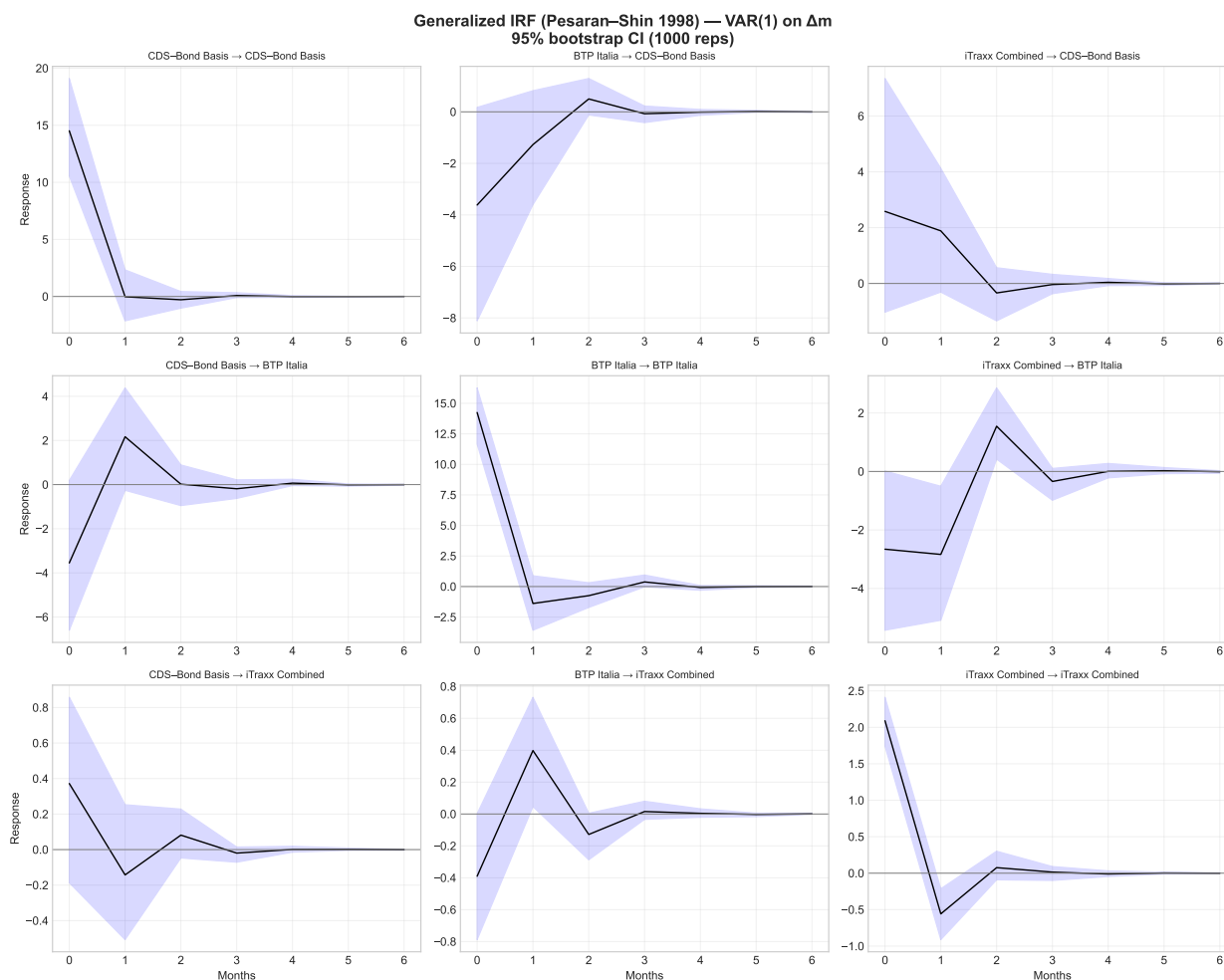


Figure 8

Generalized Impulse Response Functions. Responses of Δm to one-standard-deviation shocks. Shaded areas: 95% bootstrap confidence intervals (1000 replications).

Forecast Error Variance Decomposition. Table 17 and Figure 9 report the forecast error variance decomposition (FEVD) from the weekly VAR.

Table 17
Forecast Error Variance Decomposition (Weekly VAR)

Response $\leftarrow$ Shock	$h = 1$	$h = 4$	$h = 8$	$h = 12$
<i>CDS–Bond Basis</i>				
CDS–Bond Basis	1.000	0.986	0.985	0.985
BTP Italia	0.000	0.009	0.009	0.009
iTraxx Combined	0.000	0.005	0.006	0.006
<i>BTP Italia</i>				
CDS–Bond Basis	0.005	0.015	0.017	0.017
BTP Italia	0.995	0.977	0.974	0.974
iTraxx Combined	0.000	0.008	0.009	0.009
<i>iTraxx Combined</i>				
CDS–Bond Basis	0.040	0.042	0.043	0.043
BTP Italia	0.000	0.003	0.004	0.004
iTraxx Combined	0.960	0.954	0.953	0.953

Note: Fraction of forecast error variance of Δm_i attributable to shocks in each strategy at selected weekly horizons. Based on Cholesky decomposition of the weekly VAR. Ordering: CDS–Bond Basis, BTP Italia, iTraxx Combined.

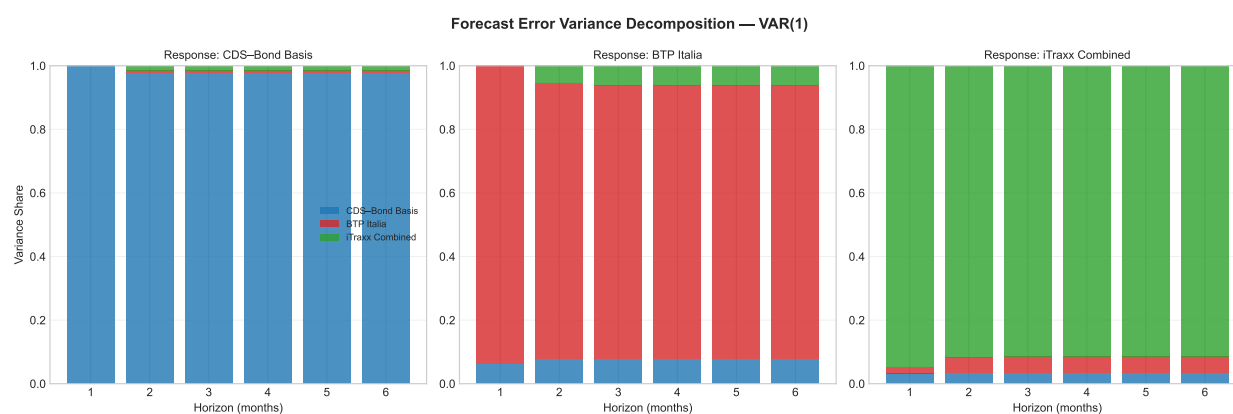


Figure 9
Forecast Error Variance Decomposition. Share of forecast error variance of Δm_i attributable to each strategy at horizons 1–12 weeks.

Threshold VAR: HIGH vs LOW Regime. Figure 10 plots GIRF separately for HIGH and LOW stress regimes, estimated from a threshold VAR.

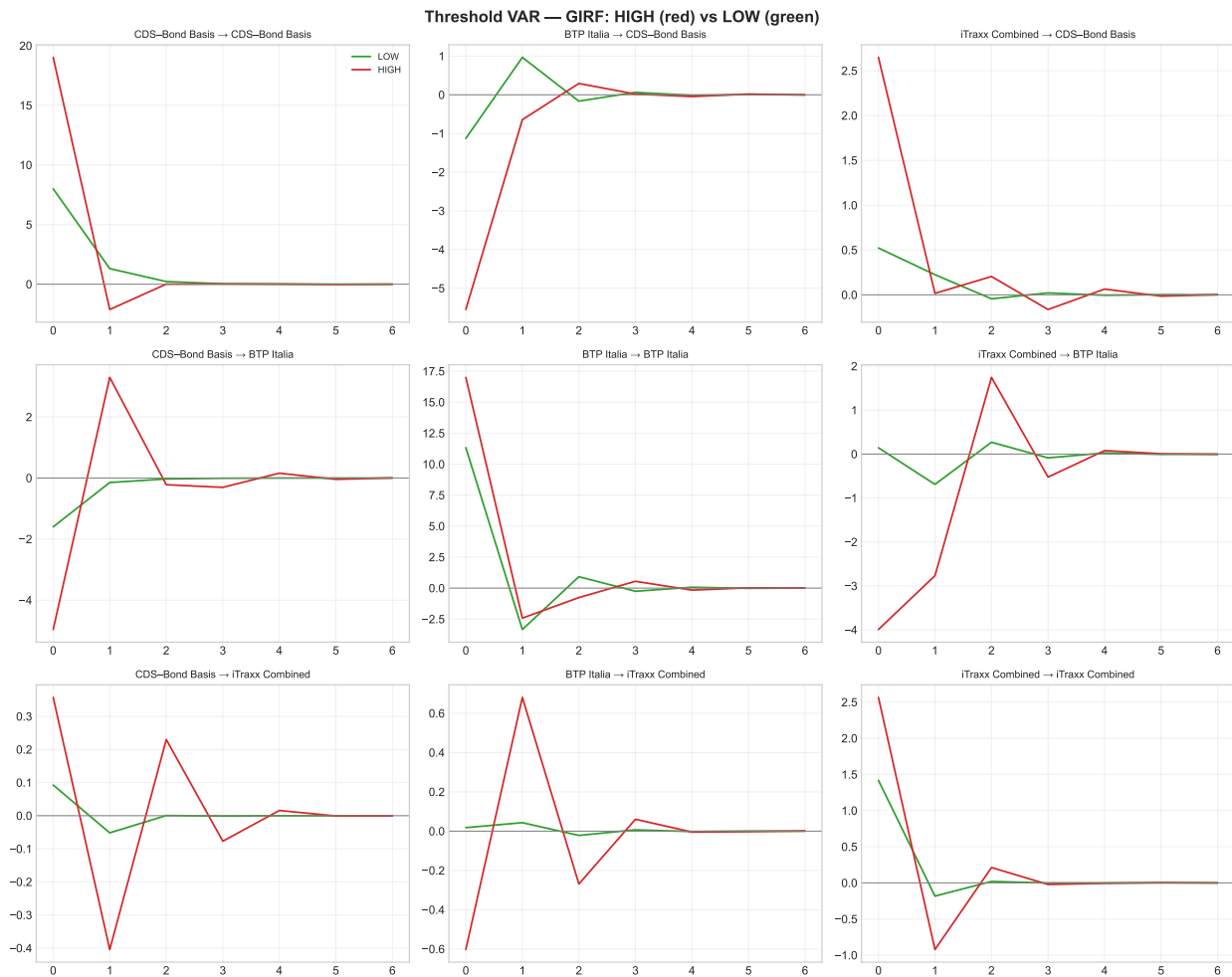


Figure 10

Threshold VAR — Generalized IRF: HIGH (red) vs. LOW (green). Responses of Δm to one-standard-deviation shocks, estimated separately in HIGH (> 120 bps) and LOW (< 60 bps) stress regimes.

6.4 Discussion: Evidence for Slow-Moving Capital

Investor Base Segmentation: The Two-Cluster Structure. CDS-Bond and iTraxx share the same institutional participants (hedge funds, dealer desks, leveraged accounts) and co-widen in stress as common funding shocks force simultaneous unwinding [Brunnermeier and Pedersen, 2009]. BTP Italia is a retail instrument with a loyalty bonus (*premio fedeltà*) that rewards hold-to-maturity behaviour, explaining why retail investors do not sell even when the basis offers apparent value relative to nominal BTPs. These preferred-habitat investors [Vayanos and Vila, 2021] do not use leverage or face margin calls, so the BTP Italia basis does not participate in institutional fire

sales—explaining the negative correlation with the corporate credit cluster that intensifies monotonically from LOW to HIGH stress.

Why Do These Arbitrages Persist? Each strategy reflects a distinct limit to arbitrage: retail investor segmentation for BTP Italia [Vayanos and Vila, 2021]; funding risk, counterparty risk, and bond illiquidity for CDS–Bond [Bai and Collin-Dufresne, 2019, Mitchell and Pulvino, 2012]; and post-crisis regulatory capital costs for the iTraxx skew, where the SLR reduces dealer ROE to $\approx 3\%$ [Boyarchenko et al., 2016, Oehmke and Zawadowski, 2017]. The persistence of alpha documented in Sections 4–5 is consistent with slow-moving capital keeping mispricings open longer than standard models predict [Duffie, 2010, Mitchell et al., 2007].

7 Conclusion

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A Appendix

A.1 BTP Italia: Instrument Details and Arbitrage Mechanics

A.1.1 Inflation Floor and Semiannual Principal Revaluation

A.1.2 Repo and Collateral Treatment

A.1.3 Comparison with US TIPS

A.2 Trade Construction Details

A.2.1 Entry/Exit Rules and Thresholds

Table 18 summarises entry and exit thresholds for each strategy. All strategies require a minimum of 6 months to maturity at entry and maintain at least one open trade after the first signal. The CDS–Bond universe is restricted to single-name CDS whose reference entity belongs to the current on-the-run iTraxx Main series; names that exit the index are no longer traded. All CDS are centrally cleared. The iTraxx skew strategy trades only on-the-run series across Main, SnrFin, SubFin, and Xover.

Table 18
Entry and Exit Thresholds by Strategy

Strategy	Entry Long	Entry Short	Exit Long	Exit Short
BTP Italia	> 40	< -50	< 10	> -10
CDS–Bond Basis	< -40	–	> 0	–
iTraxx Main	> 5	< -10	< -5	> 10
iTraxx SnrFin	> 8	< -15	< -5	> 5
iTraxx SubFin	> 10	< -20	< -10	> 10
iTraxx Xover	> 15	< -25	< -15	> 10

All thresholds in bps. CDS–Bond Basis is a one-directional negative basis trade (long bond, buy CDS protection). Minimum 6 months to maturity at entry for all strategies. Trades shorter than 3 trading days excluded ex-post.

A.2.2 P&L Calculation

A.2.3 Trade-Level Statistics

Table 19
Trade-Level Statistics by Strategy

	BTP-Italia	iTraxx <i>combined</i>	CDS-Bond Basis
Total trades	260	206	931
Long	224	146	931
Short	36	60	0
Avg duration (days)	386	225	240
Median duration (days)	212	97	119
Avg P&L per trade (%)	2.079	1.491	2.463
Median P&L per trade (%)	1.960	1.395	1.927
<i>Exit reasons (%)</i>			
Target hit	92.3	97.1	90.7
Maturity	6.9	0.5	1.0
End of sample	0.8	2.4	8.4

Note: This table reports trade-level statistics for each arbitrage strategy. P&L is the cumulative return per trade in percentage points. Duration is in calendar days. Exit reasons may not sum to 100% due to rounding.

A.2.4 Transaction Costs Transaction costs are computed at the individual trade level and applied as entry and exit fees proportional to DV01 (bonds) or duration (CDS). Fee levels are regime-dependent: at each trade date, the prevailing level of the iTraxx Main 5Y index determines which of three tiers applies (Table 20). Thresholds are set at 60 and 100 bps, reflecting the empirical widening of bid-ask spreads in stressed markets. For trades held to maturity, the exit fee is waived. All return series in Sections 3–6 are net of these costs.

Table 20
Transaction Fee Schedule (bps per unit DV01 or duration)

	LOW (Main < 60)	MID (60–100)	HIGH (Main > 100)
BTP Italia	3	5	8
CDS–Bond Basis	4	7	10
iTraxx Main	0	1	1
iTraxx SnrFin	0	1	1
iTraxx SubFin	1	2	3
iTraxx Xover	3	5	7

Fee applied as: entry DV01 $\times$ fee_bps at entry + exit DV01 $\times$ fee_bps at exit. Exit fee waived for trades held to maturity. Regime determined by iTraxx Main 5Y level at trade date: LOW (<60 bps), MID (60–100 bps), HIGH (>100 bps).

A.3 Signal-Weighted vs Equal-Weighted Comparison

Table 21 compares the signal-weighted (SW, Rebonato and Ronzani, 2021) and equal-weighted (EW, Duarte et al., 2007) indices. SW is adopted as the baseline in the main text; EW results confirm robustness to the weighting scheme.

Table 21
Summary Performance Metrics: Equal-Weighted vs Signal-Weighted

Strategy	Return (% p.a.)	Vol (% p.a.)	Sharpe	MaxDD (%)	Skew	Kurt	$\Theta_{\rho=3}$ (% p.a.)	N
Panel A: EW								
BTP-Italia	1.41	2.49	0.57	-2.99	0.54	19.76	1.32	159
CDS-Bond Basis	4.55	2.89	1.57	-4.47	2.54	53.35	4.42	203
iTraxx <i>combined</i>	1.37	2.05	0.67	-1.63	0.40	7.19	1.31	243
Panel B: SW								
BTP-Italia	1.67	2.36	0.71	-2.99	0.60	22.85	1.59	159
CDS-Bond Basis	5.37	3.15	1.71	-4.43	2.81	46.91	5.23	203
iTraxx <i>combined</i>	1.92	2.59	0.74	-2.52	0.43	5.66	1.82	243

Note: Metrics computed from daily returns (annualization factor 252). MaxDD is computed on the wealth index. $\Theta_{\rho=3}$ is the Goetzmann et al. (2007) manipulation-proof measure.

A.4 Candidate Factor Universe

A.4.1 Complete Factor List with Sources

Table 22 reports the full set of candidate risk factors used in the penalised regressions of Section 5, together with their data sources, transformations, and economic category.

Table 22
Candidate Risk Factors

Factor ID	Description	Reference
Panel A: Credit Risk Factors		
BTP_BUND	Yield differential between the Italian 10-year BTP and the German Bund.	Fausch and Sigonius [2018]
CDX_IG	First difference of the CDS spread on the CDX Investment Grade index.	Klaus and Rzepkowski [2009]
CORP_CREDIT	Broad corporate credit factor. Excess return on high-yield and leveraged loan indices over duration-matched Treasuries.	Brooks et al. [2020]
CREDIT_EU	Bond credit risk factor. Yield spread between euro-area BBB and AAA corporate bond indices.	Fama and French [1989]
CREDIT_US	Bond credit risk factor. Yield spread between Moody's BAA and AAA corporate bond indices.	Fama and French [1989]
CRED_SPR_EU	Change in the credit spread between euro-area BAA corporate bond yields and 10-year German Bund yields.	Fung and Hsieh [2001]
CRED_SPR_US	Change in the credit spread between Moody's BAA yields and 10-year U.S. Treasury yields.	Fung and Hsieh [2001]
DEF_EU	Bond default risk factor. Total return spread between long-term European corporate bonds and long-term German Bunds.	Fama and French [1993]
DEF_US	Bond default risk factor. Total return spread between long-term U.S. corporate bonds and long-term U.S. government bonds.	Fama and French [1993]
EBP	Excess Bond Premium. Residual component of aggregate corporate bond spreads after netting out expected default risk and bond characteristics via cross-sectional spread regressions; proxies time-varying credit risk compensation tied to intermediary constraints. Used with a one-month lag.	Gilchrist and Zakrajšek [2012]
EMERG_DEBT	Excess return on hard-currency emerging market debt.	Brooks et al. [2020]
EMERG_FX	Return on an equal-weighted basket of emerging market currencies versus the USD.	Brooks et al. [2020]
GLOBAL_AGG	Excess return on the Bloomberg Global Aggregate Bond Index.	Brooks et al. [2020]
ITRX_MAIN	First difference of the CDS spread on the iTraxx Europe Main index.	Klaus and Rzepkowski [2009]
MAIN_5/3_FLATTENER	First difference of the natural logarithm of the ratio between the 5-year and 3-year iTraxx Europe Main CDS spreads.	Market indicator

Note: Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 85 factors.

Table 22 (continued)

Factor ID	Description	Reference
PB_EU_CDS_1Y	Average 1-year CDS spread of European prime brokers.	Klaus and Rzepkowski [2009]
PB_EU_CDS_5Y	Average 5-year CDS spread of European prime brokers.	Klaus and Rzepkowski [2009]
PB_US_CDS_1Y	Average 1-year CDS spread of U.S. prime brokers.	Klaus and Rzepkowski [2009]
PB_US_CDS_5Y	Average 5-year CDS spread of U.S. prime brokers.	Klaus and Rzepkowski [2009]
RI_EU	Excess return on a European A-rated industrial corporate bond index.	Duarte et al. [2007]
SNR-MAIN	First difference of the 5-year CDS spread difference between the iTraxx Senior Financials index and the iTraxx Europe Main index.	Market indicator
XOVER/MAIN	First difference of the natural logarithm of the ratio between the 5-year iTraxx Crossover and 5-year iTraxx Europe Main CDS spreads.	Market indicator

Panel B: Liquidity Factors

EURIBOR_OIS	Euribor–OIS spread. Difference between 3-month Euribor and the 3-month overnight index swap rate based on EONIA or euroSTR.	Nyborg and Östberg [2014]
GC-REPO_T-BILL	Spread between the 3-month U.S. Treasury general-collateral repo rate and the 3-month U.S. Treasury bill yield.	Bai and Collin-Dufresne [2019]
HKM_IC	Intermediary capital factor. AR(1) innovation in the market-based capital ratio of primary dealers, scaled by the lagged capital ratio.	He et al. [2017]
HPW_NOISE	Treasury noise measure of market-wide illiquidity. Root mean squared deviation between observed U.S. Treasury yields and model-implied yields from a fitted smooth zero-coupon yield curve.	Hu et al. [2013]
ILLIQ	Change in the Amihud illiquidity measure. First difference of the ratio of absolute daily return to daily dollar volume.	Amihud and Mendelson [2015]
LIBOR_OIS	Libor–OIS spread. Difference between 3-month LIBOR and the 3-month overnight index swap rate.	Nyborg and Östberg [2014]
LIBOR_REPO_SHOCK	AR(2) residual of the spread between 3-month U.S. interbank LIBOR and the 3-month U.S. Treasury general-collateral repo rate.	Asness et al. [2013]
LIQNT	Pastor–Stambaugh non-traded liquidity factor. Innovation in aggregate market liquidity constructed from a cross-sectional average of stock-level order-flow-induced return-reversal measures.	Pástor and Stambaugh [2003]
LIQ_V	Value-weighted return on a 10–1 portfolio formed by sorting stocks on historical liquidity betas.	Pástor and Stambaugh [2003]
SILLIQ	Stock illiquidity shock. AR(3) residual of the aggregate Amihud illiquidity measure.	Acharya et al. [2013]
TED_SHOCK_EU	AR(2) residual of the euro TED spread (3-month Euribor minus the 3-month German government bill rate).	Asness et al. [2013]
TED_SHOCK_US	AR(2) residual of the U.S. TED spread (3-month LIBOR minus the 3-month U.S. government bill rate).	Asness et al. [2013]
ΔFAILS_PCT_TSY	First difference of the ratio of monthly total notional amount of U.S. Treasury securities fails-to-deliver reported by primary dealers, excluding TIPS, to U.S. Total Debt Outstanding.	Fleckenstein et al. [2014]

Note: Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 85 factors.

Table 22 (continued)

Factor ID	Description	Reference
Panel C: Equity Factors		
BAB_EU	Betting Against Beta (Europe). Long low-beta and short high-beta European equities, each leg rescaled to unit beta.	Frazzini and Pedersen [2014]
BAB_US	Betting Against Beta (US). Long low-beta and short high-beta U.S. equities, each leg rescaled to unit beta.	Frazzini and Pedersen [2014]
CMA_EU	Investment factor (Conservative Minus Aggressive). Return spread between European conservative and aggressive investment portfolios.	Fama and French [2017]
GMOM_EU	Global momentum factor. Return spread between recent winner and loser equity portfolios based on past 12-month returns, expressed in euros.	Asness et al. [2013]
GVAL_EU	Global value factor. Return spread between high and low book-to-market equity portfolios across developed equity regions, expressed in euros.	Asness et al. [2013]
HML_EU	Value factor (High Minus Low). Return spread between European high and low book-to-market portfolios.	Fama and French [2017]
MKT_EU	Market excess return on the European value-weighted equity market portfolio over the 1-month German government bill.	Fama and French [2017]
RB_EU	Excess return on a European bank bond index.	Duarte et al. [2007]
RMW_EU	Profitability factor (Robust Minus Weak). Return spread between European robust and weak operating profitability portfolios.	Fama and French [2017]
RS_EU	Excess return on the EuroStoxx Banks index.	Duarte et al. [2007]
SMB_EU	Size factor (Small Minus Big). Return spread between diversified European small- and large-cap portfolios.	Fama and French [2017]
UMD_EU	Momentum factor (Up Minus Down). Monthly return on a European zero-investment winners-minus-losers momentum portfolio.	Carhart [1997]
Panel D: Volatility Factors		
ATM_IV_CDX	3-month at-the-money implied volatility on options written on the CDX Investment Grade index.	Novales and Chamizo [2019]
ATM_IV_ITRX	3-month at-the-money implied volatility on options written on the iTraxx Europe Main index.	Novales and Chamizo [2019]
EP_SVIX_1M	Equity premium proxy from the SVIX methodology. Option-implied risk-neutral variance of the market simple return over the next month.	Martin [2017]
EP_SVIX_3M	Same construction as EP_SVIX_1M, using options with a 3-month horizon.	Martin [2017]
IV_BUND	3-month at-the-money implied volatility on the 10-year German Bund futures options.	Cremers et al. [2021]
IV_TSY	3-month at-the-money implied volatility on the 10-year U.S. Treasury futures options.	Cremers et al. [2021]
MOVE	Option-implied U.S. rates volatility. Yield-curve weighted average of at-the-money implied normal yield volatilities from OTC options on 2-, 5-, 10-, and 30-year constant-maturity interest rate swaps; used lagged by one month.	Bansal and Stivers [2022]

Note: Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 85 factors.

Table 22 (continued)

Factor ID	Description	Reference
PTFSBD	Return on the bond trend-following factor, constructed from lookback straddles on government bond futures.	Fung and Hsieh [2001]
PTFSCOM	Return on the commodity trend-following factor, constructed from lookback straddles on commodity futures.	Fung and Hsieh [2001]
PTFSFX	Return on the currency trend-following factor, constructed from lookback straddles on major currency futures.	Fung and Hsieh [2001]
PTFSIR	Return on the short-term interest rate trend-following factor, constructed from lookback straddles on 3-month interest rate futures.	Fung and Hsieh [2001]
PTFSSTK	Return on the equity index trend-following factor, constructed from lookback straddles on stock index futures.	Fung and Hsieh [2001]
$\Delta V2X$	Change in V2X. Monthly first difference of the V2X index, measuring 30-day implied volatility on Euro STOXX 50 options.	Chung et al. [2019]
ΔVIX	Change in VIX. Monthly first difference of the VIX index, a model-free measure of 30-day implied volatility on S&P 500 options.	Chung et al. [2019]
Panel E: Macro Factors		
5Y5Y_INFL	5-year-5-year inflation swap rate. Forward-starting EUR inflation swap rate capturing long-term inflation expectations.	Market indicator
BFCI_EU	Bloomberg Euro Area Financial Conditions Index. Composite Z-score of euro-area money, bond, and equity indicators, standardized to the pre-crisis period.	Osina [2019]
EPU_EU	Economic Policy Uncertainty index for Europe. Newspaper-based index aggregated across European countries.	Baker et al. [2016]
EPU_US	Economic Policy Uncertainty index for the United States. Newspaper-based index constructed from normalized frequency of articles referencing economy, uncertainty, and policy.	Baker et al. [2016]
ΔUF	Change in financial uncertainty. First difference of the financial uncertainty index, constructed from a panel of financial market variables.	Ludvigson et al. [2021]
ΔUM	Change in macroeconomic uncertainty. First difference of the macro uncertainty index, a model-based measure of expected volatility in the unforecastable component of a large panel of macroeconomic variables.	Ludvigson et al. [2021]
ΔUR	Change in real uncertainty. First difference of the real-activity uncertainty subindex.	Ludvigson et al. [2021]
Panel F: Interest Rate Factors		
EURUSD_3M_IV	First difference of the 3-month at-the-money implied volatility on the EUR/USD exchange rate.	Market indicator
GLOBAL_TERM	Excess return on a duration-matched global government bond index.	Brooks et al. [2020]
INFL_LINK	Excess return on global inflation-linked bonds over cash or nominal Treasuries.	Brooks et al. [2020]
R10_EU	Excess return on a European 10-year government bond portfolio.	Duarte et al. [2007]
R2_EU	Excess return on a European 2-year government bond portfolio.	Duarte et al. [2007]

Note: Factors are organized by category. Each series is reconstructed consistently with the original reference. Total: 85 factors.

Table 22 (continued)

Factor ID	Description	Reference
R5_EU	Excess return on a European 5-year government bond portfolio.	Duarte et al. [2007]
SS10Y	Difference between the 10-year EUR fixed swap par rate and the 10-year German Bund yield.	Collin-Dufresne et al. [2001]
SS2Y	Difference between the 2-year EUR fixed swap par rate and the 2-year German Schatz yield.	Collin-Dufresne et al. [2001]
SS5Y	Difference between the 5-year EUR fixed swap par rate and the 5-year German Bobl yield.	Collin-Dufresne et al. [2001]
TERM_EU	Bond term structure factor. Excess total return on long-term German Bunds over the 1-month German government bill return.	Fama and French [1993]
TERM_US	Bond term structure factor. Excess total return on long-term U.S. government bonds over the 1-month U.S. Treasury bill return.	Fama and French [1993]
YSP_EU	Yield-curve slope factor. Difference between the 5-year and 1-year German government bond yields.	Koijen et al. [2017]
YSP_US	Yield-curve slope factor. Difference between the 5-year and 1-year U.S. Treasury yields.	Koijen et al. [2017]
$\Delta 10Y_YIELD_EU$	Monthly change in the 10-year German Bund yield.	Market indicator
$\Delta 10Y_YIELD_US$	Monthly change in the 10-year U.S. Treasury yield.	Market indicator
$\Delta SLOPE_EU$	First difference of the euro yield-curve slope (10-year German Bund yield minus 3-month German government bill yield).	Ferson and Harvey [1991]
$\Delta SLOPE_US$	First difference of the U.S. yield-curve slope (10-year Treasury yield minus 3-month Treasury bill yield).	Ferson and Harvey [1991]

A.4.2 Stationarity Tests

A.5 PCA Robustness

A.5.1 Robustness across Window Length and Number of Components

Table 23 reports α and $\bar{R}^2$ for all (W, K) combinations. Results are robust to the choice of window length and number of components.

Table 23
PCA Robustness: Alpha and $\bar{R}^2$ Across Window Lengths and Number of Components

W	K	BTP Italia				CDS–Bond				Index Skew				Expl.
		Pred		Cont		Pred		Cont		Pred		Cont		Var.(%)
		α	$\bar{R}^2$	α	$\bar{R}^2$	α	$\bar{R}^2$	α	$\bar{R}^2$	α	$\bar{R}^2$	α	$\bar{R}^2$	
24	8	1.68*** (4.05)	-0.005	1.61*** (3.48)	0.054	5.43*** (5.91)	0.050	5.87*** (5.74)	0.120	2.07*** (5.38)	0.041	2.21*** (5.01)	0.049	80.8
24	11	1.65*** (3.98)	0.017	1.70*** (3.67)	0.065	5.42*** (5.60)	0.110	6.02*** (5.74)	0.121	2.10*** (5.26)	0.063	2.19*** (5.00)	0.047	88.5
24	14	1.63*** (4.10)	0.058	1.67*** (3.67)	0.071	5.30*** (5.58)	0.170	5.94*** (5.81)	0.120	2.09*** (5.27)	0.066	2.14*** (5.02)	0.127	93.6
36	8	1.63*** (3.92)	0.075	1.66*** (3.72)	0.058	5.62*** (5.74)	0.028	5.42*** (5.47)	0.093	2.06*** (4.79)	0.106	1.81*** (4.29)	0.133	76.2
36	11	1.72*** (4.06)	0.072	1.62*** (3.80)	0.042	5.72*** (5.85)	0.039	5.50*** (5.59)	0.100	2.08*** (4.91)	0.110	1.80*** (4.22)	0.141	83.5
36	14	1.65*** (3.82)	0.066	1.64*** (3.86)	0.030	5.79*** (6.05)	0.046	5.40*** (5.70)	0.117	2.13*** (5.07)	0.117	1.84*** (4.31)	0.133	88.8
48	8	1.76*** (4.25)	0.158	1.55*** (3.83)	0.066	4.95*** (4.83)	0.063	5.56*** (5.29)	0.084	1.80*** (4.43)	0.152	1.74*** (4.55)	0.265	73.2
48	11	1.82*** (4.59)	0.181	1.55*** (3.72)	0.059	4.87*** (4.74)	0.067	5.55*** (5.46)	0.079	1.78*** (4.29)	0.164	1.70*** (4.44)	0.265	80.5
48	14	1.74*** (4.08)	0.176	1.58*** (3.89)	0.063	5.12*** (5.08)	0.071	5.46*** (5.42)	0.076	1.65*** (4.11)	0.179	1.76*** (4.65)	0.269	85.7

Note: W is the rolling window length (months); K is the number of retained principal components. Expl. Var. (%) is the average cumulative variance explained by the first K PCs (time-series average across rolling windows). α is annualized (% p.a.). t -statistics in parentheses (Newey–West HAC). Pred = $PC_t \rightarrow R_{t+1}$; Cont = $PC_t \rightarrow R_t$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.5.2 Subperiod and Regime Analysis

Half-sample splits and regime-dependent alpha confirm that PCA-based alpha is stable across subperiods and increases during stress.

Table 24
PCA Subperiod Analysis (Predictive)

	Full Sample	First Half	Second Half	Stress	Normal
BTP Italia					

continued on next page

(continued)

	Full Sample	First Half	Second Half	Stress	Normal
α (% p.a.)	1.68*** (4.05)	1.48*** (2.67)	1.75*** (3.12)	4.40*** (5.28)	1.41*** (4.64)
β_{PC1}	-0.012 (-1.13)	-0.004 (-0.33)	-0.014 (-1.40)	-0.024 (-1.39)	-0.008 (-0.64)
β_{PC2}	-0.001 (-0.07)	-0.002 (-0.20)	-0.015 (-1.12)	-0.078** (-2.09)	0.001 (0.07)
β_{PC3}	-0.026 (-1.46)	-0.002 (-0.06)	-0.040* (-1.68)	-0.109** (-2.02)	-0.024 (-1.36)
β_{PC4}	-0.019 (-0.61)	0.059 (1.38)	-0.057** (-2.09)	0.048 (0.61)	-0.026 (-0.97)
β_{PC5}	-0.012 (-0.52)	-0.041 (-1.30)	0.007 (0.25)	-0.316*** (-4.00)	0.003 (0.13)
β_{PC6}	-0.028 (-1.46)	-0.043* (-1.69)	-0.021 (-1.01)	-0.075 (-0.98)	-0.023 (-1.37)
β_{PC7}	-0.001 (-0.02)	-0.052 (-1.60)	0.054 (1.60)	-0.303*** (-9.76)	0.029 (0.95)
β_{PC8}	-0.000 (-0.02)	-0.062 (-1.30)	0.038* (1.85)	-0.325*** (-4.84)	0.010 (0.47)
$\bar{R}^2$	-0.005	0.070	0.080	0.419	0.013
N	156	78	78	21	135
CDS-Bond					
α (% p.a.)	5.43*** (5.91)	7.14*** (4.85)	3.58*** (4.51)	7.93*** (3.40)	4.52*** (4.75)
β_{PC1}	0.018 (1.23)	0.033** (2.02)	-0.016 (-0.79)	0.021 (0.97)	0.003 (0.19)
β_{PC2}	0.002 (0.09)	0.014 (0.62)	-0.016 (-0.85)	0.020 (0.68)	0.002 (0.10)
β_{PC3}	0.068* (1.69)	0.073 (1.27)	0.048 (1.28)	0.092 (1.28)	0.029 (0.68)
β_{PC4}	0.059 (1.17)	-0.005 (-0.11)	0.090 (1.35)	-0.027 (-0.58)	0.091 (1.64)
β_{PC5}	-0.060** (-2.57)	-0.063 (-1.61)	-0.060* (-1.70)	-0.052 (-0.98)	-0.079** (-2.25)
β_{PC6}	0.011 (0.50)	0.030 (0.72)	-0.014 (-0.54)	0.028 (0.42)	0.005 (0.20)
β_{PC7}	-0.003 (-0.08)	0.042 (0.73)	-0.087* (-1.91)	0.053 (0.87)	-0.050 (-1.20)
β_{PC8}	0.089*** (3.03)	0.105*** (3.10)	0.024 (0.74)	0.113*** (2.78)	0.058 (1.64)
$\bar{R}^2$	0.050	0.041	0.093	0.026	0.042
N	196	98	98	50	146
Index Skew					
α (% p.a.)	2.07*** (5.38)	2.46*** (4.81)	1.49*** (2.83)	4.35*** (5.67)	1.48*** (4.50)
β_{PC1}	-0.011 (-1.20)	-0.014** (-2.25)	-0.003 (-0.16)	-0.001 (-0.09)	-0.018 (-1.44)

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(continued)

	Full Sample	First Half	Second Half	Stress	Normal
β_{PC2}	-0.009 (-0.95)	-0.008 (-1.08)	-0.009 (-0.47)	0.019 (1.26)	-0.014 (-1.28)
β_{PC3}	-0.002 (-0.13)	0.022 (1.24)	-0.033 (-1.20)	-0.005 (-0.12)	-0.007 (-0.43)
β_{PC4}	0.013 (0.78)	0.019 (1.40)	0.003 (0.11)	0.057** (1.98)	-0.005 (-0.25)
β_{PC5}	0.023 (0.89)	0.029* (1.77)	0.015 (0.32)	0.009 (0.23)	0.033 (1.11)
β_{PC6}	0.015 (0.62)	-0.010 (-0.69)	0.037 (1.33)	-0.015 (-0.44)	0.023 (0.97)
β_{PC7}	0.026 (1.47)	-0.003 (-0.18)	0.063** (2.37)	0.029 (0.58)	0.020 (1.26)
β_{PC8}	0.042** (2.11)	0.017 (0.96)	0.074*** (2.97)	-0.012 (-0.50)	0.047** (2.37)
$\bar{R}^2$	0.041	0.050	0.053	-0.063	0.094
N	208	104	104	53	155

Note: Stress regime defined as iTraxx Main 5Y > 100 bps. First/Second Half: temporal split at sample midpoint. α is annualized (% p.a.). t -statistics in parentheses (Newey–West HAC). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 25
PCA Subperiod Analysis (Contemporaneous)

	Full Sample	First Half	Second Half	Stress	Normal
BTP Italia					
α (% p.a.)	1.61*** (3.48)	1.82** (2.50)	1.38* (1.96)	5.37** (2.24)	1.28*** (4.21)
β_{PC1}	-0.025** (-1.99)	-0.034*** (-2.63)	-0.017 (-0.85)	-0.002 (-0.07)	-0.033*** (-3.54)
β_{PC2}	-0.010 (-0.97)	0.000 (0.02)	-0.018 (-1.16)	-0.005 (-0.16)	-0.013 (-1.23)
β_{PC3}	-0.028 (-1.20)	-0.041 (-1.10)	-0.025 (-0.78)	-0.153 (-1.42)	-0.016 (-0.69)
β_{PC4}	0.017 (1.05)	0.018 (0.44)	0.014 (0.62)	0.009 (0.10)	0.009 (0.66)
β_{PC5}	0.008 (0.41)	-0.026 (-1.10)	0.024 (0.86)	-0.170*** (-2.67)	0.021 (1.20)
β_{PC6}	0.003 (0.13)	-0.018 (-0.79)	0.009 (0.30)	-0.036 (-0.28)	0.000 (0.01)
β_{PC7}	-0.013 (-0.50)	-0.037 (-1.32)	0.018 (0.45)	-0.210*** (-3.01)	0.006 (0.22)
β_{PC8}	0.010 (0.60)	-0.004 (-0.21)	0.021 (0.94)	-0.092 (-1.35)	0.005 (0.25)
$\bar{R}^2$	0.054	0.026	0.026	-0.168	0.125
N	157	78	79	21	136
CDS–Bond					

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	Full Sample	First Half	Second Half	Stress	Normal
α (% p.a.)	5.87*** (5.74)	7.61*** (4.74)	4.08*** (3.86)	9.13*** (3.77)	4.84*** (4.43)
β_{PC1}	0.035*** (3.07)	0.005 (0.32)	0.051*** (3.55)	0.015 (0.92)	0.049*** (4.31)
β_{PC2}	0.010 (0.52)	0.013 (0.51)	0.003 (0.10)	0.037 (1.08)	0.008 (0.36)
β_{PC3}	0.067* (1.95)	0.074* (1.89)	0.041 (0.83)	0.076 (1.62)	0.040 (0.76)
β_{PC4}	-0.016 (-0.54)	-0.001 (-0.02)	-0.040 (-1.04)	-0.008 (-0.16)	-0.010 (-0.28)
β_{PC5}	-0.032 (-0.80)	0.022 (0.34)	-0.071** (-2.04)	0.103* (1.67)	-0.084** (-2.15)
β_{PC6}	-0.079** (-2.39)	-0.055 (-1.36)	-0.051* (-1.70)	0.016 (0.19)	-0.081** (-2.43)
β_{PC7}	0.076** (2.02)	0.108* (1.71)	0.048 (1.04)	0.028 (0.41)	0.093** (2.00)
β_{PC8}	0.051 (1.03)	0.129* (1.94)	-0.024 (-0.51)	0.075 (0.98)	0.047 (0.68)
$\bar{R}^2$	0.120	0.068	0.195	0.026	0.148
N	197	98	99	50	147
Index Skew					
α (% p.a.)	2.21*** (5.01)	2.21*** (4.60)	2.34*** (2.80)	3.38*** (4.36)	1.82*** (4.22)
β_{PC1}	0.001 (0.16)	0.005 (0.53)	-0.007 (-0.69)	-0.001 (-0.09)	0.000 (0.02)
β_{PC2}	-0.003 (-0.29)	0.003 (0.35)	0.001 (0.03)	0.011 (0.67)	-0.005 (-0.43)
β_{PC3}	0.018 (0.90)	-0.016 (-1.06)	0.053* (1.70)	0.020 (0.38)	0.019 (0.73)
β_{PC4}	0.042 (1.64)	0.003 (0.20)	0.072** (2.44)	0.024 (1.04)	0.045 (1.53)
β_{PC5}	-0.007 (-0.54)	-0.016 (-1.34)	-0.014 (-0.85)	-0.002 (-0.13)	-0.013 (-1.05)
β_{PC6}	0.012 (0.65)	-0.017 (-1.31)	0.013 (0.66)	-0.039 (-0.57)	0.020 (1.28)
β_{PC7}	-0.039** (-2.19)	-0.022 (-1.40)	-0.080*** (-3.02)	-0.053 (-0.91)	-0.042 (-1.55)
β_{PC8}	0.018 (1.21)	0.018 (0.92)	0.003 (0.12)	0.011 (0.37)	0.004 (0.25)
$\bar{R}^2$	0.049	-0.006	0.133	-0.104	0.074
N	209	104	105	53	156

Note: Stress regime defined as iTraxx Main 5Y > 100 bps. First/Second Half: temporal split at sample midpoint. α is annualized (% p.a.). t -statistics in parentheses (Newey–West HAC). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 26
PCA Alpha by Regime

Strategy	Predictive			Contemporaneous		
	Full	Stress	Normal	Full	Stress	Normal
<i>BTP Italia</i>	1.68*** (4.05)	4.40*** (5.28)	1.41*** (4.64)	1.61*** (3.48)	5.37** (2.24)	1.28*** (4.21)
<i>CDS-Bond</i>	5.43*** (5.91)	7.93*** (3.40)	4.52*** (4.75)	5.87*** (5.74)	9.13*** (3.77)	4.84*** (4.43)
<i>Index Skew</i>	2.07*** (5.38)	4.35*** (5.67)	1.48*** (4.50)	2.21*** (5.01)	3.38*** (4.36)	1.82*** (4.22)

Note: Stress regime: iTraxx Main 5Y > 100 bps. α is annualized (% p.a.). t -statistics in parentheses (Newey–West HAC). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Rolling 36-Month Alpha
(Shaded = iTraxx Main > 100 bps)

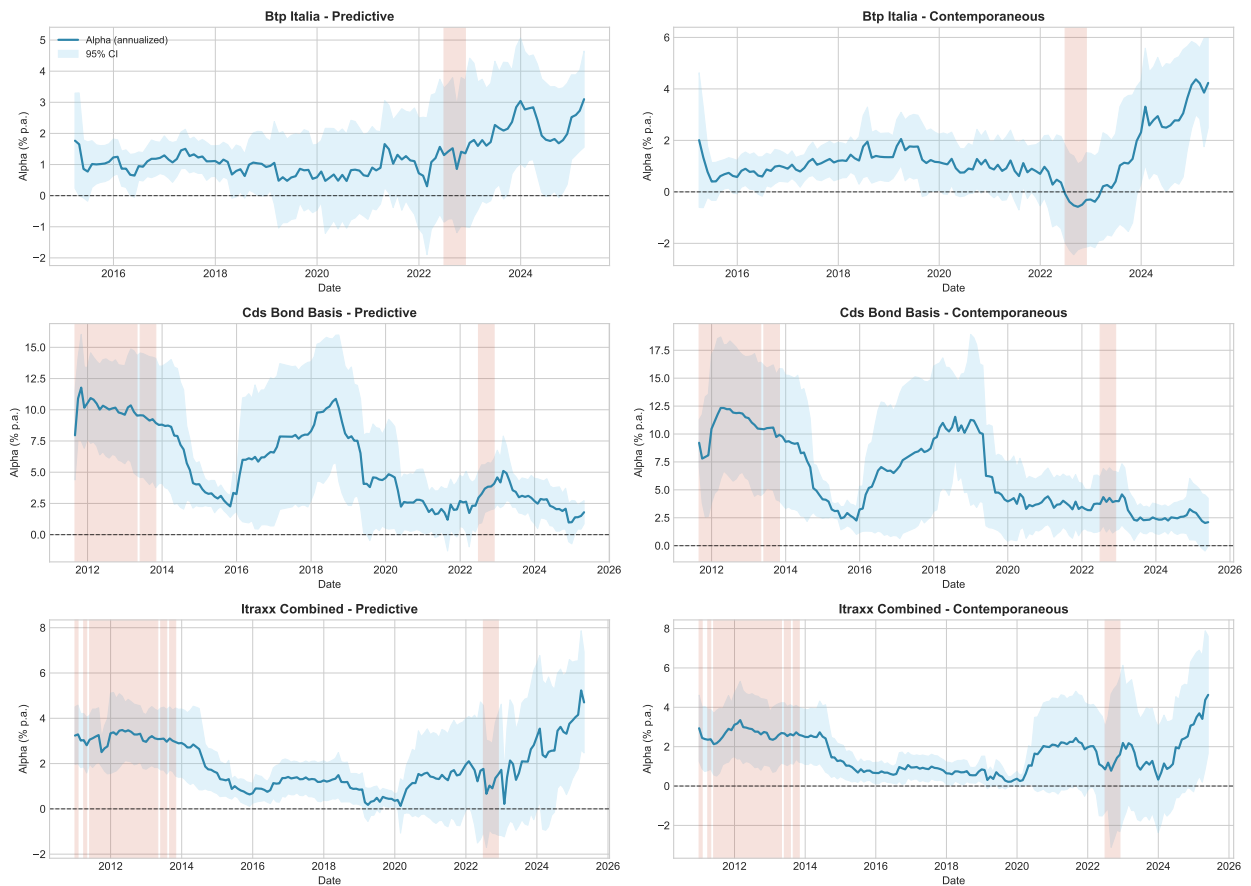


Figure 11
Rolling Alpha with Regime Shading (PCA Factors). Grey bands: 95% confidence intervals. Red shading: iTraxx Main 5Y > 100 bps.

A.5.3 Conditional Alpha: Ferson–Schadt and Threshold Robustness (PCA)

Table 27 reports Ferson–Schadt conditional alpha using PCA factors. Table 28 confirms robustness across stress thresholds (80, 100, 120 bps).

Table 27

Conditional Alpha (PCA, Contemporaneous): Ferson–Schadt (1996) Model

	BTP Italia	CDS–Bond Basis	iTraxx Combined
α_0 (ann. %)	+1.61***	+5.87***	+2.21***
α_1 (conditional)	+0.1168***	+0.2292***	+0.0736**
t -stat	(3.08)	(2.77)	(2.54)
Joint F -test	+7.06***	+3.18***	+3.36***
$\alpha(z = +1\sigma)$ ann.	+3.17%	+8.58%	+3.09%
$\alpha(z = -1\sigma)$ ann.	+0.36%	+3.08%	+1.32%
T	157	197	209

Note: Ferson and Schadt (1996) conditional model with rolling PCA factors. Timing: Contemporaneous ($PC_t \rightarrow R_{t+1}$ if predictive). z_t = standardized iTraxx Main 5Y. t -statistics in parentheses (Newey–West HAC). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 28

Conditional Alpha (PCA, Contemporaneous): Stress Threshold Robustness

Threshold	n_H	BTP Italia			CDS–Bond Basis			iTraxx Combined		
		α_1 (ann.)	t	p	α_1 (ann.)	t	p	α_1 (ann.)	t	p
80 bps	42	+1.48%	1.17	0.2424	+4.36***%	2.62	0.0087	+1.36*%	1.78	0.0750
100 bps	21	+2.37%	1.14	0.2550	+3.95%	1.63	0.1033	+1.31%	1.56	0.1198
120 bps	10	+6.77**%	2.25	0.0242	+5.16%	1.58	0.1148	+3.05**%	2.57	0.0103

Note: $D_{\text{HIGH}} = \mathbf{1}\{\text{iTraxx Main 5Y} > \text{threshold}\}$. PCA rolling factors. t -statistics (Newey–West HAC). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.5.4 Variance and Loadings Diagnostics

Figure 12 shows the time-varying cumulative variance explained.

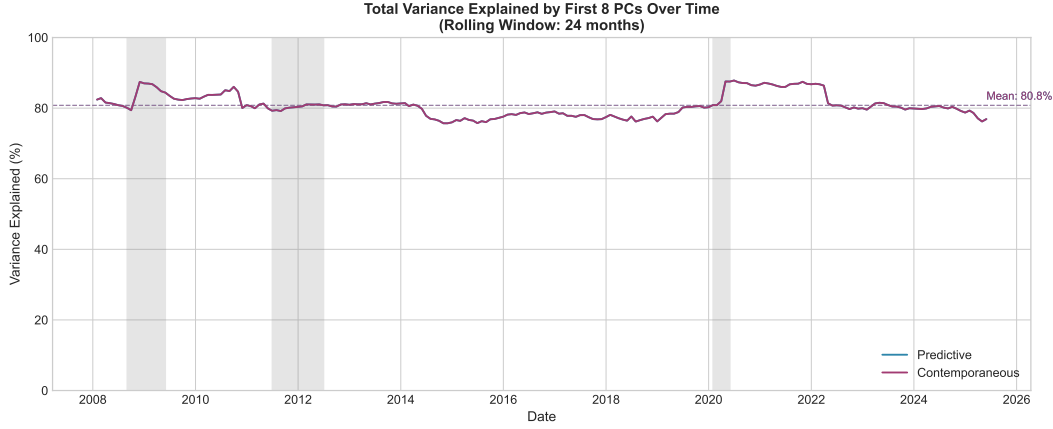


Figure 12
Time Series of Cumulative Variance Explained by Rolling PCA.

A.6 AEN Robustness

A.6.1 Sensitivity to Hyperparameters

We verify that factor selection, model dimension, and post-selection alpha are invariant to the correlation pre-cleaning threshold ($\rho \in \{0.75, 0.80, 0.85, 0.90, 0.95\}$) and the adaptive weight exponent ($\gamma \in \{0.5, 1, 2, 3\}$). In all configurations, the AEN selects the same factors with identical post-selection OLS results. No selected factor belongs to a highly correlated pair, so varying the cleaning criterion does not alter the surviving set; similarly, the Stage 1 elastic net coefficients exhibit sufficient separation between relevant and irrelevant factors to make the adaptive weighting insensitive to the exponent. Detailed results are available upon request.

A.6.2 Factor Selection Across Methods

Table 29 reports the factors selected by each penalised method for each strategy. LASSO, Elastic Net, and Adaptive LASSO produce broadly overlapping selections, while AEN—which retains only factors exceeding the 80% bootstrap stability threshold—yields the sparsest models. ALASSO-Chen selects substantially more factors across all strategies, consistent with the lighter penalisation implied by its data-driven λ_1 rule [Chen and Chen, 2008].

Table 29
Selected Factors by Method and Strategy

Method	BTP Italia	CDS–Bond Basis	iTraxx Combined
LASSO	SMB_EU, UMD_EU, ILLIQ, LIBOR_OIS, CDX_IG, SS10Y, INFL_LINK	CRED_SPR_US, ΔUF, RI_EU, PB_EU_CDS_1Y, EMERG_FX, EP_SVIX_1M, SS5Y, PTFSFX, LIBOR_OIS, SMB_EU, ΔVIX	ΔUR, YSP_EU, ΔV2X, EP_SVIX_1M, GMOM_EU, LIBOR_OIS, PB_US_CDS_5Y, SS10Y, CMA_EU, HML_EU, ΔFAILS_PCT_TSY, MOVE, LIBOR_REPO_SHOCK
Elastic Net	SMB_EU, UMD_EU, ILLIQ, LIBOR_OIS, CDX_IG, SS10Y, INFL_LINK	CRED_SPR_US, ΔUF, RI_EU, PB_EU_CDS_1Y, EMERG_FX, EP_SVIX_1M, SS5Y, PTFSFX, LIBOR_OIS, SMB_EU, ΔVIX	ΔUR, YSP_EU, ΔV2X, EP_SVIX_1M, GMOM_EU, LIBOR_OIS, PB_US_CDS_5Y, SS10Y, CMA_EU, HML_EU, ΔFAILS_PCT_TSY, MOVE, LIBOR_REPO_SHOCK

(continued)

Method	BTP Italia	CDS–Bond Basis	iTraxx Combined
Ada. LASSO	SMB_EU, UMD_EU, ILLIQ, LIBOR_OIS, CDX_IG, SS10Y, INFL_LINK	CRED_SPR_US, ΔUF, RI_EU, PB_EU_CDS_1Y, EMERG_FX, EP_SVIX_1M, SS5Y, PTFSFX	ΔUR, YSP_EU, ΔV2X, EP_SVIX_1M, GMOM_EU, LIBOR_OIS, PB_US_CDS_5Y, SS10Y, CMA_EU, HML_EU, ΔFAILS_PCT_TSY
AEN	SMB_EU, UMD_EU, ILLIQ, LIBOR_OIS, CDX_IG, SS10Y	CRED_SPR_US, ΔUF, RI_EU, PB_EU_CDS_1Y, EMERG_FX, EP_SVIX_1M, SS5Y, PTFSFX	ΔUR, YSP_EU, ΔV2X, EP_SVIX_1M, GMOM_EU, LIBOR_OIS, PB_US_CDS_5Y, SS10Y, CMA_EU, HML_EU, ΔFAILS_PCT_TSY
ALASSO-Chen	SMB_EU, UMD_EU, ILLIQ, LIBOR_OIS, TERM_EU, TERM_US, ATM_IV_CDX, ΔSLOPE_EU, Δ10Y_YIELD_US, LIBOR_REPO_SHOCK, ΔSLOPE_US, TED_SHOCK_US, ΔFAILS_PCT_TSY, DEF_US, CREDIT_EU, EPU_EU, ΔUF, YSP_US, EPU_US, MKT_EU, BAB_US, BAB_EU, PB_EU_CDS_1Y, PB_US_CDS_1Y, PB_EU_CDS_5Y, HPW_NOISE, GC-REPO-T-BILL, GMOM_EU, SILLIQ, SNR-MAIN, MAIN_5/3_FLATTENER, ITRX_MAIN, XOVER/MAIN, RMW_EU, HML_EU, SS5Y, BFCI_EU, IV_BUND, SS2Y, EURUSD_3M_IV, PTFSBD, PTFSKOM, 5Y5Y_INFL, PTFSSTK, R5_EU, RI_EU, R10_EU, GLOBAL_AGG, CORP_CREDIT, EMERG_DEBT, EMERG_FX	CRED_SPR_US, ΔUF, RI_EU, PB_EU_CDS_1Y, LIBOR_OIS, SMB_EU, CREDIT_US, DEF_US, EP_SVIX_3M, TERM_US, ΔV2X, CRED_SPR_EU, EPU_US, MKT_EU, BAB_US, TED_SHOCK_EU, PB_US_CDS_1Y, LIBOR_REPO_SHOCK, HML_EU, HPW_NOISE, CMA_EU, SS10Y, XOVER/MAIN, MOVE, EBP, PTFSBD, EURUSD_3M_IV, EURIBOR_OIS, GLOBAL_AGG	ΔUR, YSP_EU, ΔV2X, EP_SVIX_1M, GMOM_EU, LIBOR_OIS, PB_US_CDS_5Y, SS10Y, CMA_EU, HKM_IC, EP_SVIX_3M, TERM_US, DEF_EU, ΔSLOPE_EU, TED_SHOCK_EU, PB_EU_CDS_5Y, CDX_IG, SS5Y, EMERG_DEBT

Note: Each cell lists the factors selected (non-zero coefficient) by the corresponding method. Ridge excluded (selects all). AEN denotes the Adaptive Elastic Net with bootstrap stability selection ($\hat{\pi}_j \geq 60\%$).

A.6.3 Conditional Alpha: Detailed Results (AEN)

Table 30 reports the full Ferson–Schadt conditional alpha model with AEN-selected factors. Table 31 confirms robustness across stress thresholds (80, 100, 120 bps).

Table 30
Conditional Alpha: Ferson–Schadt (1996) Model

	BTP Italia	CDS–Bond Basis	iTraxx Combined
<i>Panel A: Conditional alpha</i>			
α_0 (ann. %)	+1.39***	+3.94***	+0.47
α_1 (conditional)	+0.0428	+0.2352**	+0.0471
<i>t</i> -stat	(1.31)	(2.51)	(1.55)
Joint <i>F</i> -test	3.27***	3.76***	2.17**
$\bar{R}^2$ (unconditional)	0.2173	0.2670	0.2359
$\bar{R}^2$ (conditional)	0.2778	0.2997	0.2533
$\alpha(z = +1\sigma)$ ann.	+1.67%	+7.83%	+0.87%
$\alpha(z = -1\sigma)$ ann.	+0.65%	+2.19%	-0.26%
<i>T</i>	157	197	237
<i>Panel B: Conditional betas (δ_j)</i>			
$CDX_{IG} \times z_t$	-0.002	–	–
$CRED_{SPR_US} \times z_t$	–	+0.215	–
$EMERG_{FX} \times z_t$	–	+0.016	–
$EP_{SVIX_1M} \times z_t$	–	+0.175	-0.249
$ILLIQ \times z_t$	+0.378	–	–
$LIBOR_{OIS} \times z_t$	+0.099	–	-0.379*
$PB_{EU_CDS_1Y} \times z_t$	–	-0.003	–
$PTFSFX \times z_t$	–	+0.007	–
$RI_{EU} \times z_t$	–	-0.012	–
$SMB_{EU} \times z_t$	-0.032***	–	–
$SS10Y \times z_t$	+0.006	–	–
$SS5Y \times z_t$	–	+0.015**	–
$UMD_{EU} \times z_t$	+0.017	–	–
$\Delta FAILS_{PCT_TSY} \times z_t$	–	–	+0.314*
$\Delta UF \times z_t$	–	-0.541	–
$\Delta UR \times z_t$	–	–	+1.690
$\Delta V2X \times z_t$	–	–	+0.003

Note: Ferson and Schadt (1996) conditional performance model. z_t is the standardized iTraxx Main 5Y spread (mean zero, unit variance). $\alpha_1 > 0$ indicates alpha increases with credit stress. Joint *F*-test: $H_0: \alpha_1 = \delta_1 = \dots = \delta_k = 0$. HAC (Newey–West) standard errors with 6 lags. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Factor definitions are provided in Table 22.

Table 31
Conditional Alpha: Stress Threshold Robustness

	BTP Italia			CDS–Bond Basis			iTraxx Combined		
Threshold	n_H	α_1 (ann.)	<i>t</i> -stat	n_H	α_1 (ann.)	<i>t</i> -stat	n_H	α_1 (ann.)	<i>t</i> -stat
80 bps	42	+0.67%	0.68	82	+2.38%	1.60	88	+0.48%	0.73
100 bps	21	+1.55%	0.99	50	+3.15*%	1.71	53	+0.24%	0.30
120 bps	10	+4.19%	1.40	27	+5.32*%	1.89	28	+1.37%	0.98

Note: $r_t = \alpha_0 + \alpha_1 D_{HIGH,t} + \sum_j \beta_j X_{jt} + \varepsilon_t$, where $D_{HIGH} = \mathbf{1}\{\text{iTraxx Main 5Y} > \text{threshold}\}$. n_H = number of HIGH months. Factors from bootstrap stability selection. HAC (Newey–West) standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.6.4 Rolling Alpha

Figure 13 plot rolling 36-month alpha with regime shading for AEN and PCA factors. Alpha spikes during stress episodes under both methods.

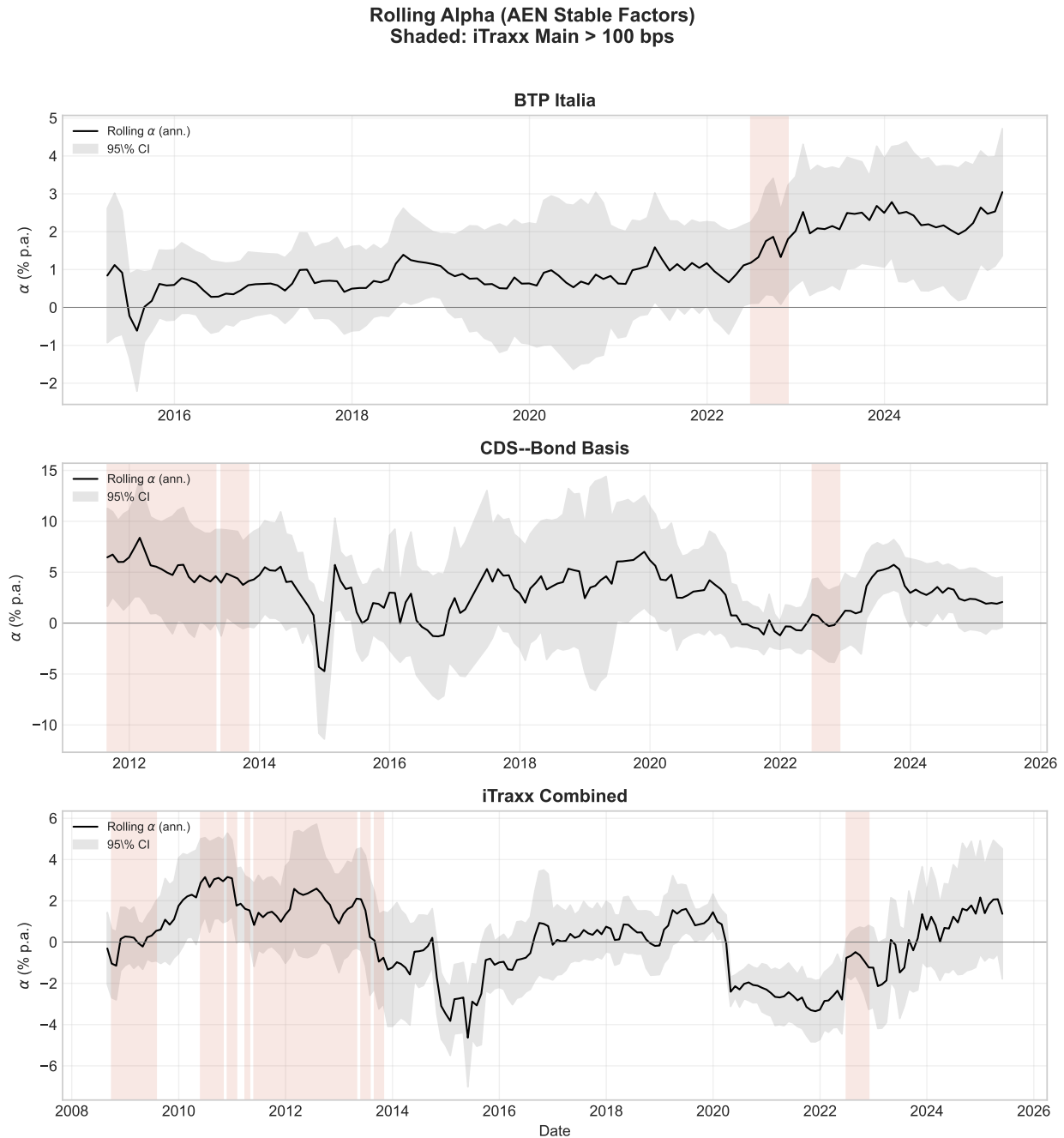


Figure 13

Rolling Alpha with Regime Shading (AEN Stable Factors). Grey bands: 95% confidence intervals. Red shading: iTraxx Main 5Y > 100 bps. Rolling window: 36 months.

A.7 Subperiod and Rolling Window Analysis

This appendix reports sub-period splits and threshold robustness for the benchmark factor models of Section 4. Each table contains Panel A (full sample, first half, second half, HIGH, and NORMAL regimes) and Panel B (stress threshold robustness at 80, 100, and 120 bps of iTraxx Main 5Y).

A.7.1 Duarte et al. (2007)

Table 32 reports sub-period splits and stress regime alpha for the Duarte et al. [2007] specification.

Table 32
Sub-Period and Regime Analysis: Duarte et al. (2007)

<i>Panel A: Sub-period alpha (% p.a.)</i>									
Period	BTP Italia			iTraxx Combined			CDS–Bond Basis		
	α	t	N	α	t	N	α	t	N
Full Sample	+1.65***	3.46	162	+2.26***	4.58	241	+5.14***	5.08	205
First Half	+1.11**	1.97	81	+2.04***	4.25	120	+7.07***	4.37	102
Second Half	+1.79**	2.35	81	+3.03***	3.64	121	+2.89***	3.49	103
HIGH	+4.74**	2.17	21	+2.71***	3.32	53	+8.63***	5.70	50
NORMAL	+1.65***	3.29	141	+1.98***	3.05	188	+3.77***	3.38	155

<i>Panel B: Alpha by stress threshold (% p.a.)</i>								
Threshold	n_H	BTP Italia		iTraxx Combined		CDS–Bond Basis		
		HIGH	NORMAL	HIGH	NORMAL	HIGH	NORMAL	
80 bps	42	+2.64	+1.30	+3.39***	+1.53	+7.30**	+3.51	
100 bps	21	+3.09	+1.46	+3.12	+2.00	+8.67**	+3.90	
120 bps	10	+5.80	+1.48	+4.24**	+1.97	+10.45**	+4.24	

Notes: Factor model: Duarte et al. (2007). EUR factors. Monthly frequency. Panel A: OLS with HAC (Newey–West) standard errors. HIGH = iTraxx Main 5Y above threshold. Panel B: $\alpha_{\text{HIGH}} = \alpha_0 + \alpha_1$ and $\alpha_{\text{NORMAL}} = \alpha_0$ from $r_t = \alpha_0 + \alpha_1 D_{\text{HIGH}} + \sum_j \beta_j X_{jt} + \varepsilon_t$. Significance stars on HIGH from t -test on α_1 . *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

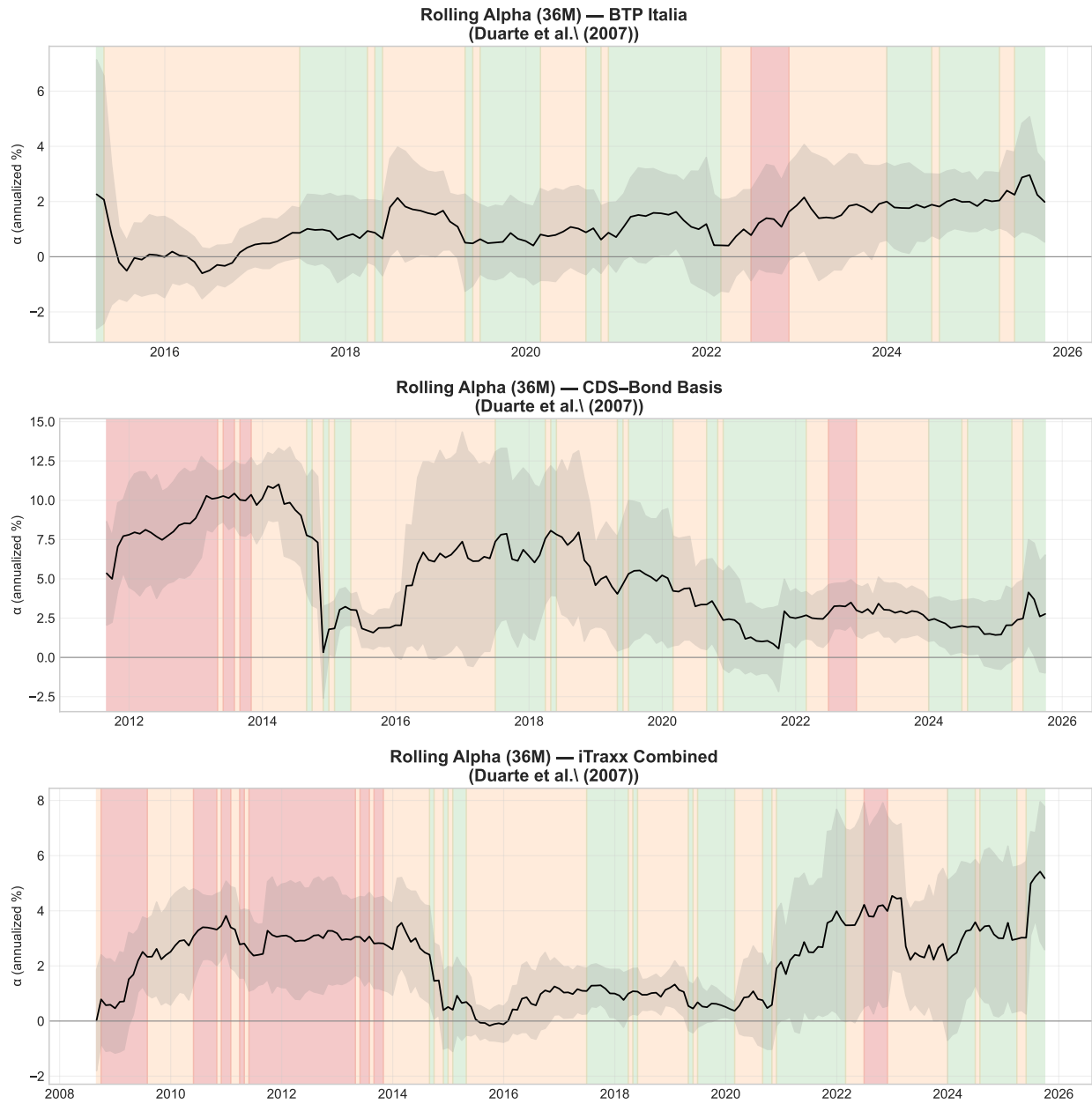


Figure 14

Rolling 36-Month Alpha with Regime Shading — Duarte et al. (2007). 95% confidence bands. Red: HIGH stress (iTraxx Main > 100 bps).

A.7.2 Brooks et al. (2020) — Active Fixed Income

Table 33 reports the same analysis under the Brooks et al. [2020] specification.

Table 33
Sub-Period and Regime Analysis: Active FI (Brooks et al. (2020))

<i>Panel A: Sub-period alpha (% p.a.)</i>									
Period	BTP Italia			iTraxx Combined			CDS–Bond Basis		
	α	t	N	α	t	N	α	t	N
Full Sample	+1.93***	3.19	163	+1.96***	4.50	243	+5.07***	5.28	207
First Half	+2.28***	2.81	81	+2.04***	4.21	121	+7.64***	4.94	103
Second Half	+1.09*	1.83	82	+2.08***	2.61	122	+2.31***	2.81	104
HIGH	+4.13**	2.15	21	+4.38***	4.48	53	+9.69***	4.80	50
NORMAL	+1.42***	3.58	142	+1.38***	3.01	190	+3.63***	4.42	157

<i>Panel B: Alpha by stress threshold (% p.a.)</i>								
Threshold	n_H	BTP Italia		iTraxx Combined		CDS–Bond Basis		
		HIGH	NORMAL	HIGH	NORMAL	HIGH	NORMAL	
80 bps	42	+3.47*	+1.44	+3.25**	+1.30	+7.94***	+3.36	
100 bps	21	+3.99	+1.73	+3.37**	+1.59	+9.22***	+3.82	
120 bps	10	+7.99**	+1.65	+5.07***	+1.59	+10.87**	+4.27	

Notes: Factor model: Active FI (Brooks et al. (2020)). EUR factors. Monthly frequency. Panel A: OLS with HAC (Newey–West) standard errors. HIGH = iTraxx Main 5Y above threshold. Panel B: $\alpha_{\text{HIGH}} = \alpha_0 + \alpha_1$ and $\alpha_{\text{NORMAL}} = \alpha_0$ from $r_t = \alpha_0 + \alpha_1 D_{\text{HIGH}} + \sum_j \beta_j X_{jt} + \varepsilon_t$. Significance stars on HIGH from t -test on α_1 . *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

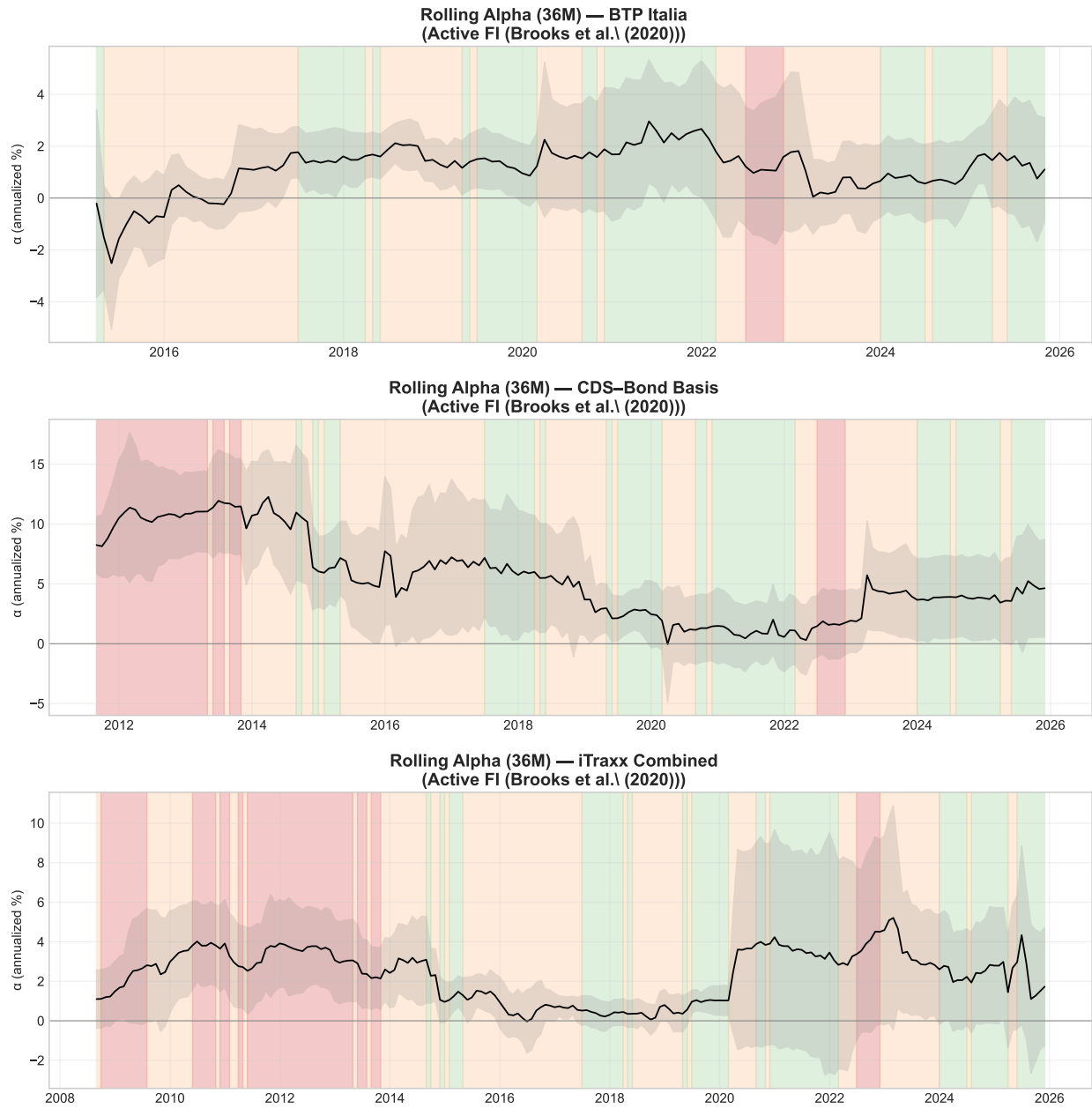


Figure 15

Rolling 36-Month Alpha with Regime Shading — Brooks et al. (2020). Same format as Figure 14.

A.7.3 Fung & Hsieh (2004)

Table 34 reports results for the Fung and Hsieh [2004] specification.

Table 34
Sub-Period and Regime Analysis: Fung & Hsieh (2001)

<i>Panel A: Sub-period alpha (% p.a.)</i>									
Period	BTP Italia			iTraxx Combined			CDS–Bond Basis		
	α	t	N	α	t	N	α	t	N
Full Sample	+1.86***	3.54	162	+3.48***	5.31	195	+4.67***	4.74	195
First Half	+2.05**	2.49	81	+5.00***	4.64	97	+7.25***	4.89	97
Second Half	+1.46**	2.21	81	+2.45***	3.88	98	+2.45***	3.59	98
HIGH	+5.62***	3.80	21	+7.25***	4.71	40	+9.56***	5.14	40
NORMAL	+1.46***	3.61	141	+2.61***	4.91	155	+2.95***	4.89	155

<i>Panel B: Alpha by stress threshold (% p.a.)</i>								
Threshold	n_H	BTP Italia		iTraxx Combined		CDS–Bond Basis		
		HIGH	NORMAL	HIGH	NORMAL	HIGH	NORMAL	
80 bps	42	+2.76	+1.52	+6.46***	+1.58	+7.49***		+2.87
100 bps	21	+4.43	+1.49	+7.21***	+2.52	+10.27**		+3.23
120 bps	10	+7.88**	+1.44	+11.71***	+2.56	+13.04**		+3.73

Notes: Factor model: Fung & Hsieh (2001). EUR factors. Monthly frequency. Panel A: OLS with HAC (Newey–West) standard errors. HIGH = iTraxx Main 5Y above threshold. Panel B: $\alpha_{\text{HIGH}} = \alpha_0 + \alpha_1$ and $\alpha_{\text{NORMAL}} = \alpha_0$ from $r_t = \alpha_0 + \alpha_1 D_{\text{HIGH}} + \sum_j \beta_j X_{jt} + \varepsilon_t$. Significance stars on HIGH from t -test on α_1 . *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

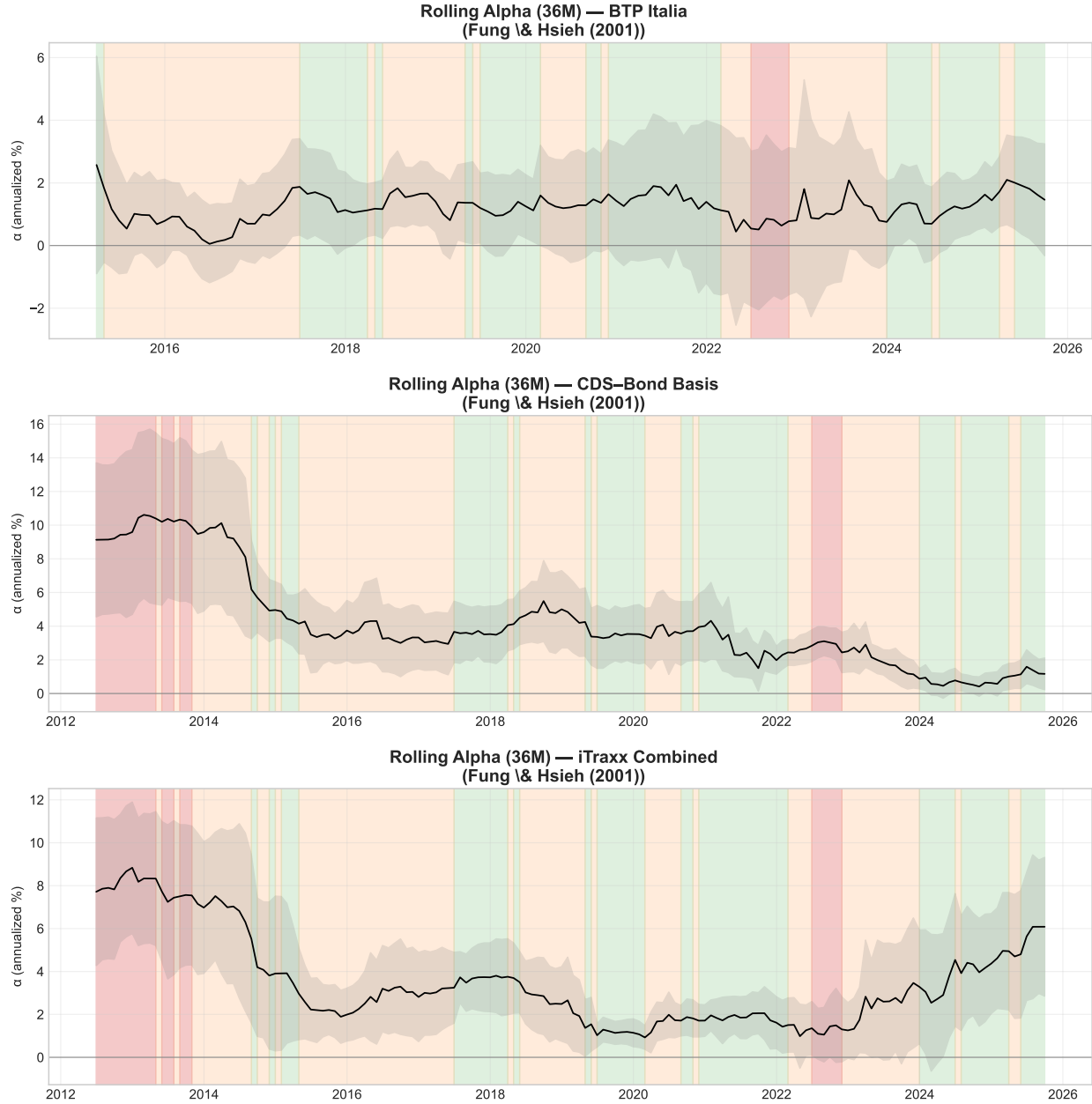


Figure 16

Rolling 36-Month Alpha with Regime Shading — Fung & Hsieh (2004). Same format as Figure 14.

A.8 Cross-Strategy Interdependencies: Additional Tests

This appendix collects robustness checks and supplementary analyses for Section 6. All tests use the same mispricing magnitudes $m_{i,t}$ and regime definitions as the main text.

A.8.1 Purged Correlations

To verify that cross-strategy co-movement is not driven solely by common risk-factor exposures, we regress each Δm_i on the AEN-selected factors from Section 5 and compute correlations on the residuals. Table 35

shows that pairwise correlations remain positive and significant after purging, indicating that the interdependencies documented in Section 6.1 reflect capital-market linkages beyond shared factor exposures.

Table 35
Correlations After Purging Common Factors

Pair	Raw	Purged	Reduction
CDS–Bond Basis vs BTP Italia	-0.260	-0.272	-4.87%
CDS–Bond Basis vs iTraxx Combined	0.182	0.130	28.87%
BTP Italia vs iTraxx Combined	-0.152	-0.093	38.69%

Note: Correlations of residuals after regressing each strategy’s Δm on common factors (EURIBOR-OIS, HKM intermediary capital, HPW noise). Reduction shows the percentage decline in absolute correlation.

A.8.2 DCC-GARCH

We estimate bivariate DCC-GARCH(1,1) models on each pair of Δm series. Figure 17 plots the time-varying conditional correlations. Spikes in conditional correlation coincide with stress episodes (COVID-19, ECB hiking cycle), corroborating the regime-dependent results in Section 6.1.

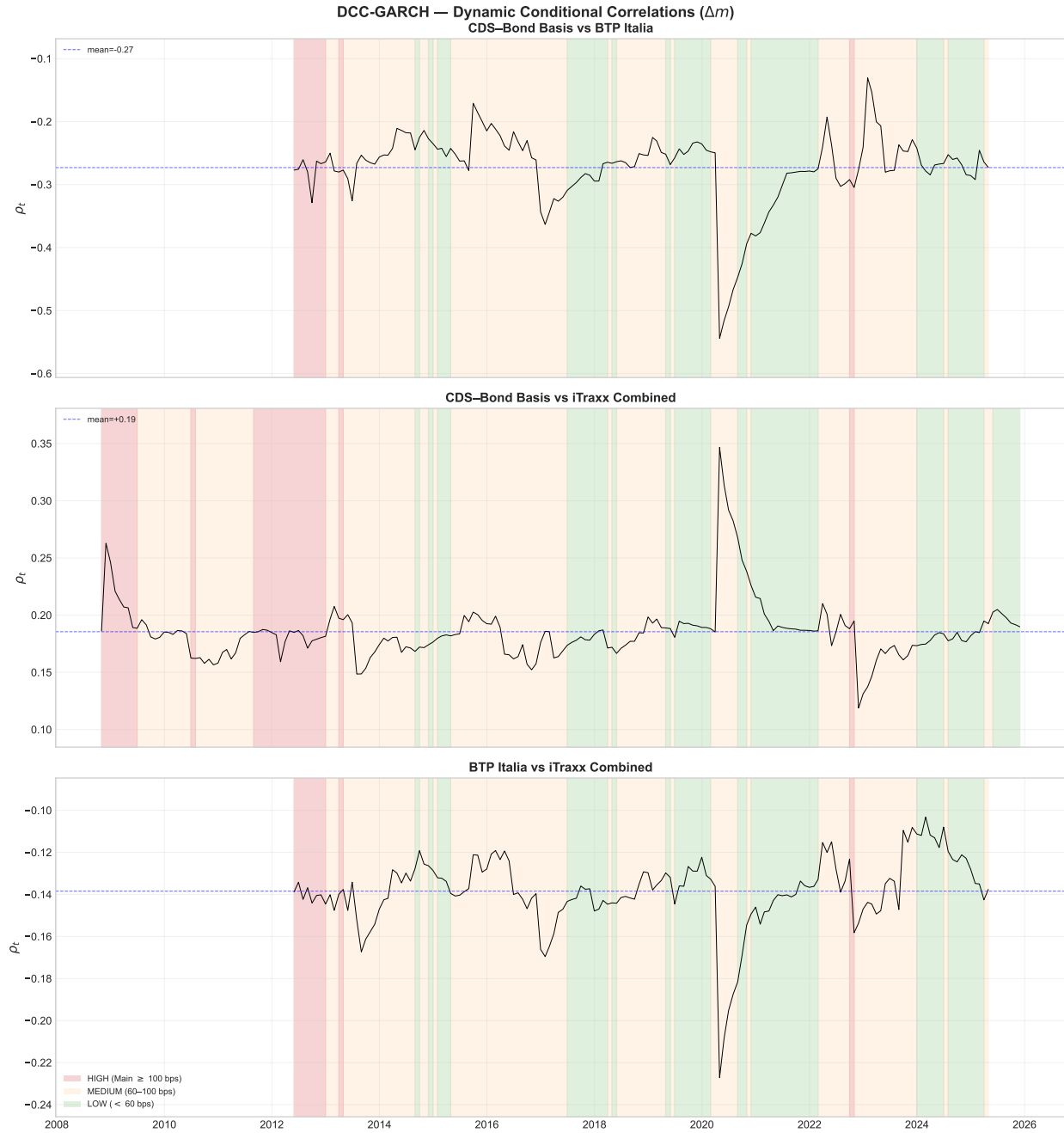


Figure 17

Dynamic Conditional Correlations (DCC-GARCH). Bivariate DCC(1,1) estimates for each pair of Δm . Shaded areas: HIGH stress regime (iTraxx Main > 100 bps).

A.8.3 Granger Causality: Lag Robustness

Table 36 reports Granger causality tests at lag lengths $p = 1, 2, 3$. The main result — iTraxx Combined Granger-causes BTP Italia — is robust across all specifications.

Table 36
Granger Causality: Robustness to Lag Length

Cause	Effect	$p = 1$		$p = 2$		$p = 3$	
		F	p -val	F	p -val	F	p -val
BTP Italia	CDS–Bond Basis	0.81	0.3696	0.37	0.6879	0.51	0.6769
iTraxx Combined	CDS–Bond Basis	2.34	0.1265	3.03**	0.0495	1.70	0.1660
CDS–Bond Basis	BTP Italia	3.97**	0.0468	1.67	0.1897	0.90	0.4414
iTraxx Combined	BTP Italia	9.51***	0.0022	4.61**	0.0104	2.99**	0.0310
CDS–Bond Basis	iTraxx Combined	0.02	0.8922	0.30	0.7434	0.29	0.8346
BTP Italia	iTraxx Combined	3.14*	0.0771	2.55*	0.0795	1.58	0.1934

Note: Granger causality F -tests from VAR(p) on Δm , estimated separately at $p = 1, 2, 3$ lags. Lag selection: HQC selects $p = 1$; AIC/FPE select $p = 2$; BIC selects $p = 0$ (forced to 1). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.8.4 Alternative Stress Proxies

Table 37 replicates key tests from Section 6 (unconditional correlations, regime correlations, interaction regressions) using iTraxx Crossover, V2X, and VIX as alternative conditioning variables instead of iTraxx Main. Results are qualitatively unchanged, confirming that the documented interdependencies are not an artifact of the stress-proxy choice.

Table 37
Robustness: Alternative Stress Proxies

Pair	iTraxx Main		iTraxx Xover		V2X		VIX	
	ρ_H	ρ_L	ρ_H	ρ_L	ρ_H	ρ_L	ρ_H	ρ_L
CDS–Bond Basis vs BTP Italia	-0.54	-0.08	-0.36	-0.10	-0.65	-0.04	-0.72	-0.02
CDS–Bond Basis vs iTraxx Combined	-0.25	-0.12	-0.09	+0.25	-0.94	-0.12	-0.08	-0.01
BTP Italia vs iTraxx Combined	+0.50	+0.20	+0.24	-0.12	+0.78	+0.17	+0.36	+0.17

Note: ρ_H and ρ_L are Pearson correlations of strategy returns in HIGH and LOW stress regimes, respectively. Regimes defined by manual thresholds on each proxy. Sign patterns are consistent across all proxies, confirming that cross-strategy interdependencies are not an artifact of the conditioning variable.

A.9 VIF Diagnostics

Tables 39–41 report Variance Inflation Factors for the EUR factor specifications used in the benchmark regressions of Section 4. VIF is computed on the factor matrix for each strategy’s sample period; because the factors are identical across strategies, differences in VIF reflect only the different estimation windows.

All equity and bond-market factors exhibit VIF below 10, confirming acceptable multicollinearity levels. The Duarte specification shows elevated VIF for the term-structure factors R2, R5, and R10 (VIF ≈ 15 –55), which is expected given the high correlation structure of yield-curve returns and does not affect inference on alpha. The Active FI and Fung & Hsieh specifications show VIF below 5 for all factors.

Table 38

Variance Inflation Factors — Duarte et al. (2007), EUR Factors

Factor	BTP Italia	CDS–Bond Basis	iTraxx Combined
Mkt-RF	5.02	5.37	5.06
SMB	1.14	1.25	1.19
HML	3.10	3.13	2.99
UMD	1.50	1.78	1.72
RS	7.25	7.96	7.90
RI	16.03	5.43	5.24
RB	11.84	2.83	2.79
R2	15.17	13.97	15.41
R5	54.39	50.88	54.08
R10	26.65	24.18	23.86

Note: VIF computed on EUR factor matrix. VIF > 10 indicates serious multicollinearity; VIF > 5 moderate concerns. Elevated VIF for R2, R5, R10 reflects the inherent correlation of Treasury return factors and does not affect inference on alpha.

Table 39

Variance Inflation Factors — Brooks et al. (2020), EUR Factors

Factor	BTP Italia	CDS–Bond Basis	iTraxx Combined
Term	6.48	6.49	6.46
Global_Term	50.36	41.58	40.74
Global_Aggregate	46.25	38.52	37.66
Inflation_Linkers	4.70	3.73	3.54
Corporate_Credit	2.49	3.51	3.55
Emerging_Debt	3.17	5.10	4.64
Emerging_Currency	1.30	1.22	1.27
UST_Volatility	1.11	1.10	1.08

Note: VIF computed on EUR factor matrix. VIF > 10 indicates serious multicollinearity; VIF > 5 moderate concerns.

Table 40

Variance Inflation Factors — Fung & Hsieh (2004), EUR Factors

Factor	BTP Italia	CDS–Bond Basis	iTraxx Combined
SNP	1.92	2.05	2.05
SIZE	1.05	1.05	1.05
PTFSBD	1.42	1.42	1.42
PTFSFX	1.45	1.44	1.44
PTFSCOM	1.19	1.22	1.22
TERM	1.03	1.01	1.01
CREDIT	1.89	1.98	1.98

Note: VIF computed on EUR factor matrix. VIF > 10 indicates serious multicollinearity; VIF > 5 moderate concerns.